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Universal equivalence of linear groups over rings

1.1.5 — “Mathematical Logic, Algebra, Number Theory, and Discrete
Mathematics”

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Introduction

Background and relevance

This dissertation belongs to model theory. In the preface to their classic book “Model Theory”, Keisler and Chang [9] describe the subject as follows:

model theory = universal algebra + mathematical logic.

Modern model theory studies the fundamental relationship between syntax, represented by a formal language, and semantics, represented by a mathematical structure described in that language. This dissertation establishes a criterion for the universal equivalence of linear groups over rings.

The study of logical properties of algebraic structures goes back to the work of A. I. Mal'cev, who proved in 1961 [17] that the elementary equivalence of the groups $G_n(K)$ and $G_m(L)$, where $G = \text{GL}, \text{SL}, \text{PGL}, \text{PSL}$, $n, m \geq 3$, and K and L are fields of characteristic 0, is equivalent to equality of their degrees $m = n$ and elementary equivalence of the fields K and L . This theorem was later extended to other types of rings and groups. These developments benefited greatly from studies of isomorphisms of linear groups over associative rings by I. Z. Golubchik, A. V. Mikhalev [12], and others. Subsequently, C. I. Beidar and A. V. Mikhalev developed [4] a general approach to proving theorems of Mal'cev type using ultraproducts and extended Mal'cev's theorem to division rings and simple associative rings.

Another way to pursue the question first posed by Mal'cev is to consider other kinds of equivalence. We can restrict the theory by allowing only one kind of quantifier and study algebraic structures within this restricted — universal — theory. The first criteria for universal equivalence were established by Yu. Sh. Gurevich and A. I. Kokorin for ordered abelian groups [13] in 1963, by N. G. Khisamiev for lattice-ordered abelian groups [14] in 1966, and then by P. C. Eklof for arbitrary abelian groups [11] in 1972. Later, other researchers obtained results on the universal equivalence of structures such as solvable groups (E. I. Timoshenko [25]), partially commutative nilpotent groups (A. A. Mishchenko and E. I. Timoshenko [18]), and partially com-

mutative metabelian Lie algebras (E. N. Poroshenko and E. I. Timoshenko [22]).

Recent results in this area include a criterion of Mal'cev type for elementary equivalence of Chevalley groups with indecomposable root systems of rank > 1 over commutative rings (with the additional assumption that they contain $1/2$ for types $\mathbf{A}_2, \mathbf{B}_l, \mathbf{C}_l, \mathbf{F}_4$, and $\mathbf{G}_2$, and $1/3$ for type $\mathbf{G}_2$), proved by E. I. Bunina [8]. The stronger property of first order rigidity has been proved both for Chevalley groups over Dedekind rings of arithmetic type and local rings (D. Segal, K. Tent [23]) and for arithmetic lattices (N. Avni, A. Lubotzky, C. Meiri [3]; M. Sohrabi, A. Myasnikov [19]).

Natural directions for further work in this area are to extend theorems of Mal'cev type to other kinds of groups or to other types of logical equivalence. The author follows the second direction, studying universal equivalence of linear groups. The paper by E. I. Bunina and the author [26] studies universal equivalence of general and special linear groups over fields. This result is closely related to, and draws on, the description of isomorphisms of linear groups in the previously mentioned work of I. Z. Golubchik and A. V. Mikhalev and in the paper by A. S. Atkarskaya, E. I. Bunina, and A. V. Mikhalev [2]. A natural continuation is the author's result for local commutative rings [27] and its extension to noncommutative local rings [28].

Aim and objectives of the dissertation

The aim of this work is to establish a criterion for universal equivalence of linear groups. To achieve this aim, the following objectives were set:

- 1) Use the method of involutions to extend the criterion for elementary equivalence of linear groups to universal equivalence of linear groups over fields of characteristic other than two.
- 2) Extend the criterion to fields of characteristic 2.
- 3) Generalise the criterion to local associative rings, both commutative and noncommutative.

Main results submitted for defence

The following results are submitted for defence:

- 1) A criterion for universal equivalence of general and special linear groups over fields of arbitrary characteristic.

- 2) A criterion for universal equivalence of general and special linear groups of degrees $n \geq 3$ over local commutative (associative with 1) rings with $1/2$.
- 3) A criterion for universal equivalence of general linear groups of degrees $n \geq 3$ over local (associative with 1) rings with $1/2$.

The dissertation establishes the following results: Theorems 1, 2, and 3.

Theorem 1. Let K and L be infinite fields. The groups $G_n(K)$ and $G_m(L)$ ($G = \text{GL}, \text{SL}, n, m \geq 2$) are universally equivalent if and only if $n = m$ and the fields K and L are universally equivalent.

Theorem 2. Let R_1 and R_2 be local commutative rings with $1/2$. The groups $G_n(R_1)$ and $G_m(R_2)$ ($G = \text{GL}, \text{SL}, n, m \geq 3$) are universally equivalent if and only if $n = m$ and the rings R_1 and R_2 are universally equivalent.

Theorem 3. Let R_1 and R_2 be local (not necessarily commutative) rings with $1/2$. The groups $\text{GL}_n(R_1)$ and $\text{GL}_m(R_2)$ ($n, m \geq 3$) are universally equivalent if and only if either $n = m$ and the rings R_1 and R_2 are universally equivalent, or $n = m$ and the rings R_1 and R_2^{op} are universally equivalent.

Originality

The results are new and were obtained by the author jointly with her supervisor (Chapter 2) and independently (Chapters 3 and 4).

Methods

The work uses methods from model theory, group theory, ring theory, and linear algebra. Chapter 2 relies substantially on concepts from linear algebra. For fields of characteristic other than 2, the classical method of involutions is used. For fields of characteristic 2, the proof is based on the study of elements of order 3. Chapters 3 and 4 also use methods from commutative algebra and the theory of associative (noncommutative) rings, together with the method of involutions.

Theoretical and practical significance

The work is theoretical. It provides insight into the relationship between the logical properties of linear groups and those of their coefficient rings. The results may find applications in group theory and model theory. The methods developed here may be useful in further studies of linear groups and in establishing criteria for other types of logical equivalence.

Validity and presentation of the results

The validity of the results is ensured by rigorous mathematical proofs. The results are consistent with those obtained by other authors in related areas.

The results were presented at the following international conferences and research seminars:

- 1) International conference “Mal’cev Readings” (Novosibirsk, 2017).
- 2) International conference “77th British Mathematical Colloquium” (Cardiff, United Kingdom, 2026).
- 3) The “Proof Theory” seminar of the Department of Mathematical Logic and the “Logic Online Seminar”, jointly with S. I. Adian’s seminar, Steklov Mathematical Institute (Moscow, 2026).
- 4) International conference “Logic Colloquium 2026 in conjunction with the British Logic Colloquium 2026” (Swansea, United Kingdom, 2026).
- 5) The “Model Theory” research seminar of the Department of Higher Algebra, Faculty of Mechanics and Mathematics, Moscow State University, led by Prof. E. I. Bunina; several presentations between 2014 and 2021.

Publications

The results of the dissertation are presented in 3 papers [26, 27, 28], all published in peer-reviewed journals.

Length and structure

The dissertation consists of an introduction, 4 chapters, a conclusion, and a bibliography. The total length is 76 pages. The bibliography contains 28 entries.

The **introduction** describes the aims of the work, outlines the history and current state of the research area, and briefly states the main results.

Chapter 1 presents the necessary background in model theory and ring theory.

The **first section** of Chapter 1 introduces the definitions of universal and existential formulas, partial isomorphism, and universal equivalence, and states a criterion for universal equivalence.

The **second section** of Chapter 1 collects the facts needed about local rings, in both the commutative and noncommutative cases.

The **third section** of Chapter 1 specifies the signatures of the structures under consideration.

Chapter 2 concerns general and special linear groups over fields. It begins by stating the main theorem for fields and then reduces the finite case to isomorphism.

The **first section** of Chapter 2 proves the easier implication: universal equivalence of fields implies universal equivalence of the groups. The proof uses the fact that all relations between matrices can be written entrywise. The same idea proves the analogous implication for more general rings in the later chapters.

The rest of the proof is divided into two cases: 1) the field has characteristic 2, and 2) the characteristic of the field is other than 2. In the second case, the standard method of involutions applies. This method does not work in characteristic 2, so it is modified by considering elements of order 3 in place of elements of order 2. In both cases, the proof follows the same outline. First, a maximal family of pairwise commuting involutions or elements of order 3 is identified. Next, elementary transvections are identified and used to identify matrices that encode addition and multiplication. The characteristic 2 case has the further complication that elements of order 3 cannot always be diagonalised. This obstacle is overcome by considering separately the case in which the base field has no nontrivial cube roots of unity: it is proved that elements of order 3 can still be put into a convenient form.

Chapter 3 extends the theorem to commutative local rings. The chapter begins with the main theorem and the restrictions on the ring and the degree under which it is proved. Here, 2 is required to be invertible, as needed for the method of involutions.

The **first section** of Chapter 3 proves the implication from rings to groups, which in the commutative case is entirely analogous to the field case. The case of finite commutative rings is also reduced to a previously known result on isomorphisms.

The **second section** of Chapter 3 applies the method of involutions directly: a maximal set of pairwise commuting involutions that are diagonal with respect to a common basis is identified. These involutions are then used to identify matrices resembling permutation matrices. In turn, these matrices make it possible to identify transvections that encode addition and multiplication in the ring.

Chapter 4 further relaxes the assumptions on the rings: they are no longer required to be commutative, but remain local. The main theorem is stated only for general linear groups. Interestingly, because the rings need not be commutative, the criterion can no longer be stated as “the groups are universally equivalent if and only if the rings are universally equivalent”. A

further alternative arises: the rings may be “antiequivalent”, meaning that one ring is universally equivalent to the opposite of the other ring. This is consistent with previously known results on isomorphisms.

Accordingly, the easier implication includes an additional case: deriving universal equivalence of the groups from universal “antiequivalence” of the rings. This statement is proved in the first section.

The treatment of the finite case is also more involved in the noncommutative setting, but it can be carried out using intermediate results for the matrix rings corresponding to the groups. This is done in the second section.

The method of involutions is then extended to the noncommutative case. The local ring assumption makes it possible to prove a result entirely analogous to the commutative case: a maximal set of pairwise commuting involutions is obtained; all of them are diagonalisable with respect to a common basis, with ± 1 on the main diagonal.

The final part of the proof shows that the form of transvections is “almost” preserved. Transvections again make it possible to write down relations in which they encode operations in the original ring or its opposite ring. This completes the proof of the theorem.

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Chapter 1

Prerequisites

1.1 Universal equivalence

Definition 1. A formula φ in the signature Σ is called *universal* if its prenex normal form is

$$\forall x_1 \dots \forall x_n \psi(x_1, \dots, x_n),$$

where ψ is a quantifier-free formula.

Definition 2. A formula φ in the signature Σ is called *existential* if its prenex normal form is

$$\exists x_1 \dots \exists x_n \psi(x_1, \dots, x_n),$$

where ψ is a quantifier-free formula.

Definition 3. Two structures $\mathfrak{A}$ and $\mathfrak{B}$ in the signature Σ are called *universally equivalent* if, for every universal sentence φ in the signature Σ ,

$$\mathfrak{A} \models \varphi \iff \mathfrak{B} \models \varphi.$$

The set of universal (existential) sentences $\{\varphi \mid \mathfrak{A} \models \varphi\}$ in the signature Σ is called the *universal (existential) theory* of the structure $\mathfrak{A}$ and is denoted by $\text{Th}_\forall(\mathfrak{A})$ ($\text{Th}_\exists(\mathfrak{A})$). Thus, $\mathfrak{A} \equiv_\forall \mathfrak{B} \iff \text{Th}_\forall(\mathfrak{A}) = \text{Th}_\forall(\mathfrak{B})$ ($\mathfrak{A} \equiv_\exists \mathfrak{B} \iff \text{Th}_\exists(\mathfrak{A}) = \text{Th}_\exists(\mathfrak{B})$).

Remark 1. By quantifier duality, $\text{Th}_\forall(\mathfrak{A}) = \text{Th}_\forall(\mathfrak{B}) \iff \text{Th}_\exists(\mathfrak{A}) = \text{Th}_\exists(\mathfrak{B})$.

Following [10], we introduce the following definition.

Definition 4. Let A be an arbitrary subset of the underlying set of a structure $\mathfrak{A}$ in the signature Σ . Then A , equipped with the induced predicates, functions, and constants, is called a *partial structure* of $\mathfrak{A}$. More precisely,

- 1) for each n -ary predicate symbol $P \in \Sigma$,

$$P_A = P_{\mathfrak{A}} \cap A^n;$$

- 2) for each n -ary function symbol $F \in \Sigma$, the operation

$$F_A(a_1, \dots, a_n)$$

is defined if and only if

$$F_{\mathfrak{A}}(a_1, \dots, a_n) \in A,$$

and in this case

$$F_A(a_1, \dots, a_n) = F_{\mathfrak{A}}(a_1, \dots, a_n);$$

- 3) for each constant symbol $c \in \Sigma$, the constant c is defined in the partial structure if and only if $c_{\mathfrak{A}} \in A$, in which case its value is $c_{\mathfrak{A}}$.

Definition 5. Let $\mathfrak{A}$ and $\mathfrak{B}$ be structures in the signature Σ , let A and B be their partial structures, and let $f: A \rightarrow B$ be a map. The map f is called a *partial isomorphism* if the following conditions hold:

- 1) f is injective;
- 2) f preserves predicates, functions, and constants:

- a) for $P^n \in \Sigma$ and $a_0, \dots, a_{n-1} \in A$,

$$P_{\mathfrak{A}}a_0 \dots a_{n-1} \Leftrightarrow P_{\mathfrak{B}}f(a_0) \dots f(a_{n-1});$$

- b) for $F^n \in \Sigma$ and $a_0, \dots, a_{n-1}, a \in A$,

$$F_{\mathfrak{A}}(a_0, \dots, a_{n-1}) = a \Leftrightarrow F_{\mathfrak{B}}(f(a_0), \dots, f(a_{n-1})) = f(a);$$

- c) for $c \in \Sigma$ and $a \in A$,

$$c_{\mathfrak{A}} = a \Leftrightarrow c_{\mathfrak{B}} = f(a).$$

Definition 6. A partial isomorphism with a finite domain is called a *finite partial isomorphism*.

Theorem 4 (*criterion for universal equivalence*, [24]). Let $\mathfrak{A}$ and $\mathfrak{B}$ be structures in the signature Σ . The structures $\mathfrak{A}$ and $\mathfrak{B}$ are universally equivalent if and only if, for every finite subsignature $\Sigma_1 \subset \Sigma$, every finite partial structure of $\mathfrak{A}$ in the signature Σ_1 is partially isomorphic to some partial structure of $\mathfrak{B}$ in the same signature, and conversely.

1.2 Local rings

Throughout this work, all rings are assumed to be associative and to have an identity distinct from zero.

Definition 7. A commutative associative ring with identity is called *local* if it has exactly one maximal ideal.

It follows from the definition that if R is a local commutative associative ring with 1 and I is its maximal ideal, then I is the set of all noninvertible elements of R . Also, if the sum of two elements of a local ring is invertible, then at least one of them is invertible.

Further information on local commutative rings can be found in [1].

For noncommutative rings, these properties are less immediate, but the following lemma holds.

Lemma 1 (see [15]). For any nonzero ring R , the following conditions are equivalent:

- 1) R has a unique maximal left ideal;
- 2) R has a unique maximal right ideal;
- 3) the quotient of R by its radical is a division ring;
- 4) the set of noninvertible elements of R is an ideal;
- 5) the set of noninvertible elements of R forms a group under addition.

Definition 8. A ring satisfying at least one, and hence all, of the conditions in the preceding lemma is called *local*.

Let R be a local ring and V a free right R -module of finite rank. By Theorem 20.13 (2) in [15], the rank of V is uniquely determined. Let $\text{GL}(V)$ be the group of invertible R -linear maps of V . Once a basis of V is fixed, $\text{GL}(V)$ can be identified with the group of invertible $n \times n$ matrices with entries in R , that is, with $\text{GL}_n(R)$, where n is the rank of V .

1.3 Conventions on signatures

We consider rings and groups in their standard signatures. For groups, the signature consists of a binary function symbol for the group operation, a unary function symbol for inversion, and a constant for the identity element: $\langle \cdot, {}^{-1}, e \rangle$. For rings, the signature consists of binary function symbols for

addition and multiplication, a unary function symbol for additive inversion, and constants for zero and one: $\langle +, \cdot, -, 0, 1 \rangle$. Unless otherwise stated, all signatures under consideration include equality.

Chapter 2

General and special linear groups over fields

This chapter is devoted to the proof of the following theorem:

Theorem 1. Let K and L be infinite fields. The groups $G_n(K)$ and $G_m(L)$ ($G = GL, SL, n, m \geq 2$) are universally equivalent if and only if $n = m$ and the fields K and L are universally equivalent.

Remark 2. Universal equivalence of fields and linear groups does not coincide with elementary equivalence. Indeed, by Mal'cev's theorem, it suffices to give an example of two fields that are universally equivalent but not elementarily equivalent. Consider the algebraic closure of the field of rational numbers $\overline{\mathbb{Q}}$ and the rational function field $\overline{\mathbb{Q}}(t)$ over it. These fields are not elementarily equivalent, since one is algebraically closed, whereas the other contains elements with no square root, such as t . Nevertheless, they are universally equivalent. To see this, consider the tower of extensions

$$\overline{\mathbb{Q}} \subset \overline{\mathbb{Q}}(t) \subset \overline{\overline{\mathbb{Q}}(t)}.$$

First, $\overline{\mathbb{Q}} \equiv \overline{\overline{\mathbb{Q}}(t)}$, since both are algebraically closed fields of characteristic 0. Second, their existential theories satisfy the natural inclusions

$$\text{Th}_{\exists}(\overline{\mathbb{Q}}) \subset \text{Th}_{\exists}(\overline{\mathbb{Q}}(t)) \subset \text{Th}_{\exists}(\overline{\overline{\mathbb{Q}}(t)}),$$

but elementary equivalence implies that the existential theories of the first and last fields coincide. Thus all three existential theories coincide, and $\overline{\mathbb{Q}} \equiv_{\forall} \overline{\mathbb{Q}}(t)$.

Remark 3. If the fields K and L are finite, universal equivalence coincides with isomorphism, and we can refer to the following theorem presented in the lectures [20]:

Theorem 5. Let n and m be positive integers, $n, m \geq 2$, and let K and L be arbitrary fields. Then the following conditions are equivalent:

- 1) $n = m$ and $K \cong L$,
- 2) $\text{GL}_n(K) \cong \text{GL}_m(L)$,
- 3) $\text{SL}_n(K) \cong \text{SL}_m(L)$.

2.1 Universal equivalence of fields implies universal equivalence of linear groups

In this section we prove the easier implication of the theorem.

Proposition 1. Let K and L be universally equivalent infinite fields. Then, for every positive integer n ,

$$\text{GL}_n(K) \equiv_{\forall} \text{GL}_n(L), \text{SL}_n(K) \equiv_{\forall} \text{SL}_n(L).$$

Proof. We describe the linear groups as subsets of K^{n^2} equipped with the indicated operations.

$$\begin{aligned} \text{GL}_n(K) &= \left\{ (a_1, \dots, a_{n^2}) \in K^{n^2} \mid \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^n a_{(i-1)n+\sigma(i)} \neq 0 \right\}; \\ \text{SL}_n(K) &= \left\{ (a_1, \dots, a_{n^2}) \in K^{n^2} \mid \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^n a_{(i-1)n+\sigma(i)} = 1 \right\}. \end{aligned}$$

Multiplication in $\text{GL}_n(K)$ and $\text{SL}_n(K)$ is given by the following formula:

$$(a_1, \dots, a_{n^2}) \cdot (b_1, \dots, b_{n^2}) = (c_1, \dots, c_{n^2}),$$

where

$$c_k = c_{(i-1)n+j} = \sum_{l=1}^n a_{(i-1)n+l} \cdot b_{(l-1)n+j}, k = 1, \dots, n^2, i, j = 1, \dots, n.$$

Thus any finite partial structure of $\text{GL}_n(K)$ or $\text{SL}_n(K)$ can be described as a finite partial structure of K^{n^2} . Since the fields K and L are universally equivalent, we can find a finite partial structure of L^{n^2} , partially isomorphic to it. Consequently, for the original partial structure of $\text{GL}_n(K)$ or $\text{SL}_n(K)$, we can find a finite partial structure of $\text{GL}_n(L)$ or $\text{SL}_n(L)$, respectively, partially isomorphic to it, and conversely.

This proves the proposition. $\square$

We divide the proof of the converse implication into two cases: 1) the fields K and L have characteristic two; 2) the fields K and L have characteristics other than two.

2.2 Fields of characteristic 2

In this part of the work we prove the following theorem.

Theorem 6. Let K and L be infinite fields of characteristic 2, and suppose that the groups $G_n(K)$ and $G_m(L)$ ($G = \text{GL}, \text{SL}$, $n, m \geq 2$) are universally equivalent. Then $n = m$ and the fields K and L are universally equivalent.

Throughout this section, all fields under consideration have characteristic 2.

Lemma 2. Let K be a field of characteristic 2, and let $n = 2$ or $n = 3$. Suppose that $A \in \text{GL}_n(K)$ ($A \in \text{SL}_n(K)$), $A \neq I$, and $A^2 = I$. Then, with respect to some basis, the matrix A has the form $I + E_{12}$.

Proof. Let $V_0 = \text{Ker}(A + I)$, a nonzero subspace of K^n . For every vector $t \in V_0$, we have $At = t$. Since A is not the identity operator, $\dim V_0 < n$.

Consider a vector $x \notin V_0$. If $(A + I)x = \lambda x$, then $(A + I)^2x = \lambda^2x = 0$. Since $x \neq 0$, it follows that $\lambda = 0$, and hence $x \in V_0$, a contradiction.

Therefore, $(A + I)x = y$, where x and y are linearly independent vectors. The same equality gives $Ax = x + y$. The action of A on y is given by $Ay = A(Ax + x) = x + Ax = y$. It follows immediately that, for $n = 2$, the matrix of A with respect to the basis $\{y, x\}$ is $I + E_{12}$.

If $n = 3$ and $\dim V_0 = 2$, the assertion holds. Indeed, with respect to a basis consisting of the vectors y, x and a vector in V_0 linearly independent of y , the matrix of A has the required form. Now suppose that $\dim V_0 = 1$. Extend the linearly independent set $\{y, x\}$ to a basis by adding a vector z , and consider the action of A on z : $Az = z + h$, where z and h are linearly independent. Then $Ah = A(Az + z) = z + Az = h$, so $h \in V_0$ and $h = \mu y$. Consider the sum $Az + \mu Ax = z + \mu y + \mu x + \mu y = z + \mu x$. It follows that $z + \mu x \in V_0$, so $z + \mu x = \nu y$, contrary to the linear independence of x, y, z . This proves the assertion. $\square$

We next consider two cases: the field contains nontrivial cube roots of unity, i.e. there are two elements $\xi_1 \neq \xi_2$, distinct from $\xi_0 = 1$, such that $\xi_1^3 = \xi_2^3 = 1$; or, conversely, the field contains no such roots.

2.2.1 Properties of linear groups expressible by an existential formula when the field contains nontrivial cube roots of unity

Lemma 3. Suppose that the field K of characteristic 2 contains nontrivial cube roots of unity. Then $\mathrm{GL}_n(K)$ has a set of $3^n - 1$ pairwise commuting matrices of order 3 and no larger such set; likewise, $\mathrm{SL}_n(K)$ has a set of $3^{n-1} - 1$ pairwise commuting matrices of order 3 and no larger such set. With respect to some basis, all these matrices have the form $\mathrm{diag} [\xi_1, \dots, \xi_n]$, where all ξ_i are cube roots of unity.

Proof. The assertion follows from the fact that each such matrix is diagonalisable and that a set of pairwise commuting matrices has a common eigenvector. Thus all these matrices are diagonal with respect to a common basis. $\square$

Let $\mathcal{A}(\mathrm{GL}_n(K))$ ($\mathcal{A}(\mathrm{SL}_n(K))$), or simply $\mathcal{A}$ when the group is clear from the context) denote a fixed maximal set of matrices of the form $\mathrm{diag} [\xi_1, \dots, \xi_n]$, whose existence was established in Lemma 3.

Lemma 4. Suppose that the field K contains nontrivial cube roots of unity and $n \geq 4$. Then, in the groups $\mathrm{GL}_n(K)$ and $\mathrm{SL}_n(K)$, a matrix of order 2 whose conjugates by all matrices in $\mathcal{A}$ consist of exactly three distinct pairwise commuting matrices has the form $I + E_{ij}$ ($i \neq j$) with respect to a basis relative to which the matrices of the system $\mathcal{A}$ retain their form.

Proof. Suppose that a matrix A satisfies $A \neq I$ and $A^2 = I$, and that conjugation of A by all matrices in $\mathcal{A}$ gives exactly three distinct matrices. Assume that A has at least two nonzero off-diagonal entries a_{pq} and a_{ts} such that $(p, q) \neq (s, t)$. Conjugation can then multiply a_{pq} and a_{ts} independently by any cube root of unity, i.e. it gives more than three distinct conjugates, a contradiction. Now suppose that $(p, q) = (s, t)$ and there are no other

nonzero off-diagonal entries. Then

$$A = \begin{pmatrix} 1 & 0 & \dots\dots\dots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & 1 & \vdots \\ \vdots & & & a_{ii} & 0 & \dots & 0 & a_{ij} & \vdots \\ \vdots & & & 0 & 1 & & 0 & \vdots \\ \vdots & & & \vdots & & \ddots & \vdots & \vdots \\ \vdots & & & 0 & & & 1 & 0 & \vdots \\ \vdots & & & a_{ji} & 0 & \dots & 0 & a_{jj} & \vdots \\ \vdots & & & & & & & 1 & \vdots \\ \vdots & & & & & & & & \ddots & 0 \\ 0 & \dots\dots\dots & & 0 & 1 \end{pmatrix}.$$

Thus we need to consider a block of the form

$$\begin{pmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{pmatrix}.$$

Conjugating it by the diagonal matrices of the system $\mathcal{A}$ gives three matrices:

$$\begin{pmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{pmatrix}, \begin{pmatrix} a_{ii} & \xi^2 a_{ij} \\ \xi a_{ji} & a_{jj} \end{pmatrix}, \begin{pmatrix} a_{ii} & \xi a_{ij} \\ \xi^2 a_{ji} & a_{jj} \end{pmatrix},$$

where ξ is a nontrivial cube root of 1.

Since any 2 of the three matrices above commute, we obtain the following relation:

$$\begin{aligned} \begin{pmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{pmatrix} \begin{pmatrix} a_{ii} & \xi a_{ij} \\ \xi^2 a_{ji} & a_{jj} \end{pmatrix} &= \begin{pmatrix} a_{ii}^2 + \xi^2 a_{ij} a_{ji} & \xi a_{ii} a_{ij} + a_{ij} a_{jj} \\ a_{ii} a_{ji} + \xi^2 a_{ji} a_{jj} & \xi a_{ij} a_{ji} + a_{jj}^2 \end{pmatrix} = \\ &= \begin{pmatrix} a_{ii} & \xi a_{ij} \\ \xi^2 a_{ji} & a_{jj} \end{pmatrix} \begin{pmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{pmatrix} = \begin{pmatrix} a_{ii}^2 + \xi a_{ij} a_{ji} & a_{ii} a_{ij} + \xi a_{ij} a_{jj} \\ \xi^2 a_{ii} a_{ji} + a_{ji} a_{jj} & \xi^2 a_{ij} a_{ji} + a_{jj}^2 \end{pmatrix}, \end{aligned}$$

which implies, in particular, that $a_{ij} a_{ji} = 0$. Using the condition $A^2 = I$, we obtain $a_{ii} = a_{jj} = 1$.

Suppose that $a_{ij} \neq 0$. If $a_{ij} \neq 1$, make the change of basis given by the matrix $I + (a_{ij} + 1)E_{jj}$, which commutes with every matrix in the system $\mathcal{A}$:

$$\begin{pmatrix} 1 & 0 \\ 0 & a_{ij} \end{pmatrix} \begin{pmatrix} 1 & a_{ij} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{a_{ij}} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

as required. $\square$

Lemma 5. Suppose that the field K contains nontrivial cube roots of unity and $n \geq 4$. Then, in the groups $\mathrm{GL}_n(K)$ and $\mathrm{SL}_n(K)$, a maximal set of matrices of order 2 with pairwise distinct centralisers, such that conjugating each of them by all matrices in $\mathcal{A}$ gives exactly three distinct pairwise commuting matrices, consists of $n^2 - n$ matrices of the form $I + c_{ij}E_{ij}$ ($i, j = 1, \dots, n; i \neq j, c_{ij} \in K^*$) with respect to a basis relative to which the matrices of the system $\mathcal{A}$ retain their form.

Proof. The proof of the preceding lemma shows that all the matrices of order 2 under consideration have the required form with respect to a common basis (relative to which the matrices of the system $\mathcal{A}$ retain their form). Since their centralisers are pairwise distinct, there are at most $n^2 - n$ such matrices, and this number is attained. $\square$

Lemma 6. Suppose that the fields K and L of characteristic 2 contain nontrivial cube roots of unity and that $\mathrm{G}_n(K) \equiv_{\forall} \mathrm{G}_m(L)$ ($\mathrm{G} = \mathrm{GL}$ or SL). Then $n = m$.

Proof. Suppose that $\mathrm{GL}_n(K) \equiv_{\forall} \mathrm{GL}_m(L)$. Consider the existential formula

$$\begin{aligned} \exists A_1 \dots \exists A_{3^{N-1}-1} \left(\bigwedge_{i=1}^{3^{N-1}-1} (\neg A_i = I) \& \bigwedge_{\substack{i,j=1 \\ i \neq j}}^{3^{N-1}-1} (\neg A_i = A_j) \& \right. \\ \left. \& \bigwedge_{i,j=1}^{3^{N-1}-1} (A_i A_j = A_j A_i) \& \bigwedge_{i=1}^{3^{N-1}-1} (A_i^3 = I) \right), \quad (2.1) \end{aligned}$$

which asserts the existence of a set of $3^{N-1} - 1$ pairwise distinct nonidentity pairwise commuting matrices of order 3. The largest value of N for which formula (2.1) holds in both groups is, by Lemma 3, the same and equals $N = n + 1 = m + 1$.

Similarly, if $\mathrm{SL}_n(K) \equiv_{\forall} \mathrm{SL}_m(L)$, then the largest N for which formula (2.1) holds in both groups satisfies $N = m = n$. $\square$

2.2.2 Properties of linear groups expressible by an existential formula when the field contains no nontrivial cube roots of unity

Lemma 7. If the field K of characteristic 2 contains no nontrivial cube roots of unity, then every matrix of order three in $\mathrm{GL}_n(K)$ and $\mathrm{SL}_n(K)$ has, with respect to some basis, a block-diagonal form in which each block is one of

the matrices $E_{11} + E_{12} + E_{21}$, $E_{12} + E_{21} + E_{22}$, or I . A maximal set of pairwise commuting matrices of order 3 consists of $3^{\lfloor n/2 \rfloor} - 1$ matrices that, with respect to some basis, have a block-diagonal form in which each block is one of the matrices $E_{11} + E_{12} + E_{21}$, $E_{12} + E_{21} + E_{22}$, or I .

Proof. We use the method described in [7]. Consider a nonidentity matrix A such that $A^3 = I$. Let $\sigma = I + A + A^2$. Note that $\sigma^2 = \sigma$. The space K^n is the sum of the subspaces $V_0 = \sigma K^n$ and $V_1 = (I + \sigma)K^n$, since every vector $x \in K^n$ can be written as $x = \sigma x + (I + \sigma)x$.

On the subspace V_0 , the operator A acts as the identity. Indeed, if $x \in V_0$, then $x = \sigma y$ for some $y \in K^n$, and hence

$$Ax = A\sigma y = A(I + A + A^2)y = \sigma y = x.$$

We now show that $V_1 = \text{Ker } \sigma$. If $x \in V_1$, then $x = (I + \sigma)y$ for some $y \in K^n$. Moreover,

$$\sigma x = \sigma(I + \sigma)y = (\sigma + \sigma^2)y = 0.$$

For $x \in \text{Ker } \sigma$, we clearly have $x = (I + \sigma)x \in V_1$.

It also follows that the sum of V_0 and V_1 is direct: $K^n = V_0 \oplus V_1$.

Consider the action of A on the subspace V_1 . First suppose that $x \in V_1$ is an eigenvector of A . Then $Ax = \lambda x$, so $\lambda^3 = 1$, and hence $\lambda = 1$. This gives a contradiction: $0 = \sigma x = (I + A + A^2)x = x + x + x = x$.

Now let $x \in V_1$ be a nonzero vector that is not an eigenvector of A . Then $y = Ax \in V_1$ ($x = (I + \sigma)t$ for some $t \in K^n$, $y = Ax = A(I + \sigma)t = (I + \sigma)(At) \in V_1$), and x and y are linearly independent. The action of A on y is given by $Ay = A^2x = (\sigma + A + I)x = y + x$. Thus the restriction of A to the invariant subspace $\langle x, Ax \rangle$ has the form

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

with respect to the basis consisting of x and $y = Ax$.

We now show that the subspace V_1 is a direct sum of two-dimensional invariant subspaces of the same form. Proceeding by induction, suppose that we have already constructed a direct sum of subspaces $\langle x_1, Ax_1 \rangle, \dots, \langle x_k, Ax_k \rangle$ (which is invariant under A) that is not all of V_1 , and that there are vectors outside the linear span $\langle x_1, Ax_1, \dots, x_k, Ax_k \rangle$ that are not eigenvectors. Choose such a vector x_{k+1} and show that Ax_{k+1} is linearly independent of all the preceding vectors. Suppose otherwise: $Ax_{k+1} = \lambda x_{k+1} + v$, where $v \in \langle x_1, Ax_1, \dots, x_k, Ax_k \rangle$. On the quotient space $V_1 / \langle x_1, Ax_1, \dots, x_k, Ax_k \rangle$, we have $\bar{A}\bar{x}_{k+1} = \lambda\bar{x}_{k+1}$. Since $\bar{A}^2 + \bar{A} + I = 0$, it follows that $(\lambda^2 + \lambda + 1)\bar{x}_{k+1} = 0$.

By the choice of x_{k+1} , its class $\bar{x}_{k+1}$ is nonzero, so $\lambda^2 + \lambda + 1 = 0$. Therefore λ is a nontrivial cube root of unity, a contradiction.

It remains to show that all the matrices have the required form with respect to a common basis. Let ξ be a nontrivial cube root of unity. Then the distinct cube roots of unity are $1, \xi, \xi^2$. Over the algebraic closure of K , and even over the extension $K(\xi)$, all the matrices under consideration are simultaneously diagonalisable, with $1, \xi, \xi^2$ on the diagonal. We already know that over K they have a block form and that the characteristic polynomial of each nonidentity block is $x^2 + x + 1$. Hence the eigenvalues of the nonidentity blocks are ξ and ξ^2 , so the diagonal matrices over $K(\xi)$ contain equal numbers of ξ and ξ^2 .

We now show that we can pair ξ and ξ^2 so that they occupy consecutive positions in all the matrices simultaneously. Let A, B be commuting matrices of order 3 with entries in K , and let v be a common eigenvector corresponding to nonidentity eigenvalues λ_A and λ_B , respectively (over $K(\xi)$):

$$Av = \lambda_A v, \quad Bv = \lambda_B v.$$

Let φ be the automorphism of the extension $K(\xi)$ over K that interchanges ξ and ξ^2 . Applying it entrywise to the relations above gives:

$$A\varphi(v) = \varphi(\lambda_A)\varphi(v), \quad B\varphi(v) = \varphi(\lambda_B)\varphi(v),$$

Thus, along with the eigenvector v , the matrices A and B have the eigenvector $\varphi(v)$ with eigenvalues conjugate to the original ones. Over K , this pair of eigenvalues corresponds to 2×2 blocks with eigenvalues ξ and ξ^2 in the same positions in A and B .

If these 2×2 blocks coincide in diagonal form over $K(\xi)$, they also correspond to the same block over K . If their diagonal forms are $\text{diag}[\xi, \xi^2]$ and $\text{diag}[\xi^2, \xi]$, and the first corresponds to the block $E_{11} + E_{12} + E_{21}$ over K , then the second corresponds over K to its inverse, the block $E_{12} + E_{21} + E_{22}$. $\square$

Let $\mathcal{B}(\text{GL}_n(K))$ ($\mathcal{B}(\text{SL}_n(K))$) be a fixed maximal set of pairwise commuting matrices of order 3 from Lemma 7.

Lemma 8. Suppose that the infinite field K contains no nontrivial cube roots of unity and $n > 5$. Then, in $\text{GL}_n(K)$ and $\text{SL}_n(K)$, a matrix that has the form $I + E_{12}$ with respect to some basis can be characterised by an existential formula.

Proof. The matrices in $\mathcal{B}(\text{GL}_n(K))$ ($\mathcal{B}(\text{SL}_n(K))$) can be characterised in the respective groups by an existential formula: there exist $3^{\lfloor \frac{n}{2} \rfloor} - 1$ distinct pairwise commuting matrices of order 3. This set of matrices splits into

conjugacy classes consisting of matrices with the same number of nonidentity blocks. Assign to each class the number of nonidentity blocks in each of its matrices. Thus there are $\left\lfloor \frac{n}{2} \right\rfloor$ conjugacy classes, and the k th class contains $2^k C_{\lfloor n/2 \rfloor}^k$ matrices, $k = 1, \dots, \left\lfloor \frac{n}{2} \right\rfloor$.

Number the matrices consecutively across all conjugacy classes, in descending order of the class numbers, and include the corresponding pairwise conjugacy relations for the matrices in each class in our formula. The resulting formula remains existential.

We show that there is no bijection of the conjugacy classes that does not fix the first class. It suffices to prove that

$$2 \left\lfloor \frac{n}{2} \right\rfloor < 2^k C_{\lfloor n/2 \rfloor}^k \text{ for } 1 < k \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Since $n > 5$, we have $\left\lfloor \frac{n}{2} \right\rfloor > 2$. Thus

$$\text{for } k = \left\lfloor \frac{n}{2} \right\rfloor \text{ we obtain } 2 \left\lfloor \frac{n}{2} \right\rfloor < 1 \cdot 2^{\lfloor n/2 \rfloor},$$

$$\text{and for } 1 < k < \left\lfloor \frac{n}{2} \right\rfloor \text{ we obtain } 2 \left\lfloor \frac{n}{2} \right\rfloor < 2^k \cdot \left\lfloor \frac{n}{2} \right\rfloor \leq 2^k \cdot C_{\lfloor n/2 \rfloor}^k.$$

Thus we can identify all matrices with exactly one nonidentity block. These matrices split into pairs of inverses. Choose one matrix from each pair, denote them by $A_1, \dots, A_{\lfloor n/2 \rfloor}$, and let $\mathcal{C} = \{A_1, \dots, A_{\lfloor n/2 \rfloor}\}$. We may assume that the blocks of size 2 occupy consecutive positions and that the index of each matrix equals the position of its nonidentity block.

Consider a matrix $B = (b_{ij})$ that commutes with A_1 , and show that $b_{1i} = b_{2i} = 0$, $b_{i1} = b_{i2} = 0$, $i = 3, \dots, n$. Without loss of generality, we may assume that the nonidentity block of A_1 is $E_{11} + E_{12} + E_{21}$. Then

$$\begin{aligned} \begin{pmatrix} 1 & 1 & & & \\ 1 & 0 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ b_{31} & b_{32} & b_{33} & \dots & b_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \dots & b_{nn} \end{pmatrix} = \\ = \begin{pmatrix} b_{11} + b_{21} & b_{12} + b_{22} & b_{13} + b_{23} & \dots & b_{1n} + b_{2n} \\ b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{31} & b_{32} & b_{33} & \dots & b_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \dots & b_{nn} \end{pmatrix}, \end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ b_{31} & b_{32} & b_{33} & \dots & b_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \dots & b_{nn} \end{pmatrix} \begin{pmatrix} 1 & 1 & & & \\ 1 & 0 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix} = \\
& = \begin{pmatrix} b_{11} + b_{12} & b_{11} & b_{13} & \dots & b_{1n} \\ b_{21} + b_{22} & b_{21} & b_{23} & \dots & b_{2n} \\ b_{31} + b_{32} & b_{31} & b_{33} & \dots & b_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{n1} + b_{n2} & b_{n1} & b_{n3} & \dots & b_{nn} \end{pmatrix}.
\end{aligned}$$

It follows that $b_{1i} = b_{1i} + b_{2i} = b_{2i} = 0$, $b_{i1} = b_{i1} + b_{i2} = b_{i2} = 0$, $i = 3, \dots, n$.

Similarly, if $B = (b_{ij})$ commutes with A_m , then $b_{2m-1,i} = b_{2m,i} = 0$, $b_{i,2m-1} = b_{i,2m} = 0$, $i = 1, \dots, 2\widehat{m}-1, 2\widehat{m}, \dots, n$.

Now consider a matrix C that commutes with $\lfloor \frac{n}{2} \rfloor - 1$ of the matrices A_i (we may assume that these are $A_1, \dots, A_{\lfloor n/2 \rfloor - 1}$, making a change of basis that permutes the blocks if necessary) and has order 2. By the above, C has a block-diagonal structure: it has either $\lfloor \frac{n}{2} \rfloor = \frac{n}{2}$ blocks of size 2, or $\lfloor \frac{n}{2} \rfloor - 1$ blocks of size 2 and one final block of size 3.

Moreover, every block of C squares to I , and all but the last commute with a block of order 3. Thus, for a block commuting with a block of order 3, we have:

$$\begin{aligned}
& \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a+b & a \\ c+d & c \end{pmatrix}, \\
& \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a+c & b+d \\ a & b \end{pmatrix}, \\
& \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.
\end{aligned}$$

The matrix relations imply that $b = c$ and $b(a + d) = 0$. If $b = 0$, then $a = d = 1$ and $b = c = 0$. Otherwise, $a = d$; since $a = b + d$, we get $b = 0$, hence also $c = 0$, and again $a = d = 1$.

We have shown that all blocks of C except the last are identity blocks, while the last has size 2 or 3 and order 2. Hence, by Lemma 2, with respect to some basis (relative to which the matrices of the system $\mathcal{B}$ will, in general, change their form), the matrix C has the form $I + E_{12}$. $\square$

Lemma 9. Suppose that the infinite field K contains no nontrivial cube roots of unity and $n > 5$. If n is even, then, in the groups $\text{GL}_n(K)$ and $\text{SL}_n(K)$, for each matrix of the system $\mathcal{C}$ (defined in the proof of Lemma 8), there are

no two distinct commuting matrices of order 2 with distinct centralisers that commute with every matrix of the system $\mathcal{C}$ except the given one. If n is odd, then, in the groups $\text{GL}_n(K)$ and $\text{SL}_n(K)$, for a matrix of the system $\mathcal{C}$, there exist 2 distinct commuting matrices of order 2 with distinct centralisers that commute with every matrix of the system $\mathcal{C}$ except the given one.

Proof. If n is even, then, by the proof of Lemma 8, a matrix A_e of order 2 commuting with all but one (denoted by B_e , with only its first block assumed to be nonidentity) of the matrices of the system $\mathcal{C}$ has, with respect to some basis, the form below. With respect to this basis, all matrices of the system $\mathcal{C}$ except possibly B_e retain their form, and only the first block of B_e may change:

$$A_e = \begin{pmatrix} 1 & 1 & & & \\ 0 & 1 & & & \\ & & I & & \\ & & & \ddots & \\ & & & & I \end{pmatrix},$$

where all identity blocks have size 2, and the block decomposition agrees with that of the matrices in the system $\mathcal{C}$. All matrices of order 2 that commute with every matrix in $\mathcal{C}$ except B_e , and also with A_e , have the following form with respect to the same basis:

$$\begin{pmatrix} 1 & \alpha & & \\ 0 & 1 & & \\ & & I & \end{pmatrix},$$

but the centraliser of every such matrix coincides with the centraliser of A_e .

Now let n be odd. By Lemma 8, a matrix A_o of order 2 that commutes with all but one (denoted by B_o) of the matrices of the system $\mathcal{C}$ has the following form with respect to the same basis:

$$A_o = \begin{pmatrix} \tilde{A}_o & & & \\ & I & & \\ & & \ddots & \\ & & & I \end{pmatrix},$$

where the block $\tilde{A}_o$ has size 3, all identity blocks have size 2, and the block decomposition agrees with that of the matrices in the system $\mathcal{C}$. As two commuting matrices of order 2 with distinct centralisers that commute with every matrix of the system $\mathcal{C}$ except B_o , we may take the following matrices

(with respect to the same basis):

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

This proves the lemma. $\square$

Before proving an analogue of Lemma 6 for fields containing no nontrivial cube roots of unity, we make a few remarks on existential formulas that hold in the groups $\text{GL}_n(K)$ and $\text{SL}_n(K)$ for $2 \leq n \leq 5$.

For $n = 2$, in the groups $\text{GL}_n(K)$ and $\text{SL}_n(K)$, a maximal set of pairwise commuting matrices of order 3 consists of two matrices. A matrix of order 2 has the following property: every matrix of order 2 that commutes with it has the same centraliser.

For $n = 3$, in the groups $\text{GL}_n(K)$ and $\text{SL}_n(K)$, a maximal set of pairwise commuting matrices of order 3 also consists of two matrices, but there exist 2 commuting matrices of order 2 with distinct centralisers.

For $n = 4$ and $n = 5$, in the groups $\text{GL}_n(K)$ and $\text{SL}_n(K)$, a maximal set of pairwise commuting matrices of order 3 consists of 8 matrices. We show how to distinguish the degrees in this case. Note that a matrix which, with respect to some basis, is a Jordan block of maximal size with eigenvalue 1 has the largest order of the form 2^k among all matrices in the group (there may be matrices with a different Jordan form and the same order; for example, Jordan blocks of sizes 3 and 4 both have order 4). Thus, for $n = 4$, the largest k such that the group contains an element of order 2^k is 2. As already noted, the Jordan block below is an example of a matrix of order 4:

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

whereas for $n = 5$, the largest order of the form 2^k is 8, the order of the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Lemma 10. Suppose that the infinite fields K and L of characteristic 2 contain no nontrivial cube roots of unity and that the groups $G_n(K)$ and $G_m(L)$ are universally equivalent ($G = GL$ or SL). Then $n = m$.

Proof. Consider the formula

$$\begin{aligned} \exists A_1 \dots \exists A_{3^N-1} \left(\bigwedge_{i=1}^{3^N-1} (\neg A_i = I) \& \bigwedge_{\substack{i,j=1 \\ i \neq j}}^{3^N-1} (\neg A_i = A_j) \& \right. \\ \left. \& \bigwedge_{i,j=1}^{3^N-1} (A_i A_j = A_j A_i) \& \bigwedge_{i=1}^{3^N-1} (A_i^3 = I) \right). \end{aligned}$$

It asserts the existence of a set of $3^N - 1$ distinct pairwise commuting matrices of order 3. Since the linear groups are universally equivalent, the largest N for which the formula holds in $G_n(K)$ and $G_m(L)$ ($G = GL, SL$) is the same, and, by Lemma 7, $N = \lceil \frac{n}{2} \rceil = \lceil \frac{m}{2} \rceil$.

If $N = 1$, then the formula

$$\begin{aligned} \exists A_1 \exists A_2 \exists C ((\neg A_1 = A_2) \& (\neg A_1 = I) \& (\neg A_2 = I) \& \\ \& (A_1 A_2 = A_2 A_1) \& (A_1^2 = I) \& (A_2^2 = I) \& (A_1 C = C A_1) \& (\neg A_2 C = C A_2)), \end{aligned}$$

which asserts the existence of two distinct commuting matrices of order 2 with distinct centralisers, is either true in both groups or false in both. If it is true, then $n = m = 3$; otherwise, $n = m = 2$.

If $N = 2$, then the largest k for which the formula

$$\exists J \left(\bigwedge_{i=1}^{2^k-1} (\neg J^i = I) \& (J^{2^k} = I) \right),$$

holds in both groups is the same. If the largest value of k is 2, then $n = m = 4$; if $k = 3$, then $n = m = 5$.

Now suppose that $N \geq 3$. Then $n, m \geq 6$, so we can use the methods

from Lemmas 8 and 9. Consider the formula

$$\begin{aligned}
& \exists A_{11} \dots \exists A_{1,2C_M^1} \exists A_{21} \dots \exists A_{2,2^2 C_M^2} \dots \exists A_{M,1} \dots \exists A_{M,2^M C_M^M} \\
& \quad \exists B_1 \exists B_2 \exists C \exists C_{k,i,j} (k = 1, \dots, M; i, j = 1, \dots, 2^k C_M^k) \\
& \quad \left(\bigwedge_{k=1}^M \bigwedge_{i=1}^{2^k C_M^k} (\neg A_{k,i} = I) \& \bigwedge_{k=1}^M \bigwedge_{i=1}^{2^k C_M^k} \bigwedge_{l=1}^M \bigwedge_{\substack{j=1 \\ (k,i) \neq (l,j)}}^{2^l C_M^l} (\neg A_{k,i} = A_{l,j}) \& \right. \\
& \quad \& \bigwedge_{k=1}^M \bigwedge_{i=1}^{2^k C_M^k} \bigwedge_{l=1}^M \bigwedge_{j=1}^{2^l C_M^l} (A_{k,i} A_{l,j} = A_{l,j} A_{k,i}) \& \bigwedge_{k=1}^M \bigwedge_{i=1}^{2^k C_M^k} (A_{k,i}^3 = I) \& \\
& \quad \& \bigwedge_{k=1}^M \bigwedge_{i=1}^{2^k C_M^k} \bigwedge_{j=1}^{2^k C_M^k} (A_{k,i} = C_{k,i,j}^{-1} A_{k,j} C_{k,i,j}) \& \bigwedge_{i=1}^{C_M^1} (A_{1,i} A_{1,i+C_M^1} = I) \& \\
& \quad \& (\neg B_1 = I) \& (\neg B_2 = I) \& \bigwedge_{i=1}^{C_M^1-1} (B_1 A_{1,i} = A_{1,i} B_1) \& \bigwedge_{i=1}^{C_M^1-1} (B_2 A_{1,i} = A_{1,i} B_2) \& \\
& \quad \left. \& (B_1^2 = I) \& (B_2^2 = I) \& (B_1 B_2 = B_2 B_1) \& (B_1 C = C B_1) \& (\neg B_2 C = C B_2) \right).
\end{aligned}$$

When M is the integer part of half the degree of the group, this formula identifies, among the matrices of the system $\mathcal{B}$, the conjugacy class consisting of matrices with exactly one nonidentity block (these are $A_{11}, \dots, A_{1,2C_M^1}$), and the system $\mathcal{C}$ within that class, $\mathcal{C} = \{A_{11}, \dots, A_{1,C_M^1}\}$. It then asserts the existence of two commuting matrices of order 2 with distinct centralisers that commute with every matrix of the system $\mathcal{C}$ except A_{1,C_M^1} . This sentence is true or false simultaneously in both groups. If it is true in both groups for $N = M$, then $n = m = 2M + 1$; otherwise, $n = m = 2M$. $\square$

2.2.3 Properties expressible by an existential formula without assuming the existence of nontrivial cube roots of unity

Proposition 2. Let K and L be infinite fields of characteristic 2, and suppose that $G_n(K) \equiv_{\forall} G_m(L)$ ($G = \text{GL}$ or SL). Then $n = m$.

Proof. Consider the following two formulas:

$$\exists A_1 \dots \exists A_{3^N-1} \left(\bigwedge_{i=1}^{3^N-1} (\neg A_i = I) \& \bigwedge_{\substack{i,j=1 \\ i \neq j}}^{3^N-1} (\neg A_i = A_j) \& \right. \\ \left. \bigwedge_{i,j=1}^{3^N-1} (A_i A_j = A_j A_i) \& \bigwedge_{i=1}^{3^N-1} (A_i^3 = I) \right), \quad (2.2)$$

$$\exists J \left(\bigwedge_{i=1}^{2^k-1} (\neg J^i = I) \& (J^{2^k} = I) \right). \quad (2.3)$$

Since the groups are universally equivalent, the largest N for which formula (2.2) holds is the same for both groups. Thus, for general linear groups, the possible degrees are $N, 2N, 2N+1$, while for special linear groups they are $N+1, 2N, 2N+1$.

The largest k for which formula (2.3) holds is also the same. A Jordan block of maximal size with eigenvalue 1 has order 2^k , where k is the smallest positive integer such that $2^k \geq n, 2^k \geq m$. Hence:

$$2^{k-1} < n \leq 2^k, \\ 2^{k-1} < m \leq 2^k.$$

Thus the following cases are possible for general linear groups:

1) exactly one of the numbers $N, 2N, 2N+1$ belongs to the interval $(2^{k-1}, 2^k]$; then $n = m$ equals that number, and we also determine whether the fields K and L contain nontrivial cube roots of unity;

2) $2N, 2N+1 \in (2^{k-1}, 2^k]$; then the fields K and L contain no nontrivial cube roots of unity, so we can apply Lemma 10 to conclude that $m = n$.

For special linear groups, in addition to the analogous cases (exactly one of the numbers $N+1, 2N, 2N+1$ belongs to the interval $(2^{k-1}, 2^k]$, or $2N, 2N+1 \in (2^{k-1}, 2^k]$), we must consider the case $N+1, 2N \in (2^{k-1}, 2^k]$. In this case, $N = 2^{k-1}$ and $N+1$ is the size of the smallest Jordan block of order 2^k . In the group $\text{SL}_{N+1}(K)$, a Jordan block of size $N+1$ has a commutative centraliser. If $N = 1$, both groups have degree $N+1 = 2N = 2$. If $N > 1$ and the groups contain a matrix of order 2^k with a noncommutative centraliser, then their degree is $2N$; otherwise, their degree is $N+1$. Indeed, for $N = 2$, the group of degree $2N = 4$ contains the following two noncommuting matrices:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

which commute with the Jordan block

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For $N > 2$, the difference $2N - (N + 1) \geq 2$, and an example of two noncommuting matrices in the centraliser of a Jordan block of order 2^k is also easy to construct. $\square$

Definition 9. A finite partial isomorphism

$$\Phi: G_n(K) \rightarrow G_m(L) \quad (G = GL, SL)$$

is called *special* if $\text{dom}(\Phi)$ contains the following finite set of matrices: the identity matrix, a maximal set of pairwise commuting matrices of order 3 (the system $\mathcal{A}$ or $\mathcal{B}$, depending on whether the field contains nontrivial cube roots of unity), all matrices of the form $I + E_{ij}$ ($i, j = 1, \dots, n; i \neq j$) with respect to the corresponding basis, and the auxiliary matrices from the formulas used in the proofs of Lemma 10 and Proposition 2, including inverse matrices and all intermediate products occurring in the relations (there are finitely many of these).

Lemma 11. Let K and L be infinite fields of characteristic 2, and suppose that $G_n(K) \equiv_{\forall} G_m(L)$ ($G = GL, SL$). Then every special finite partial isomorphism

$$\Phi: G_n(K) \rightarrow G_m(L) \quad (G = GL, SL)$$

maps the matrix $I + E_{12}$ to a matrix that has the form $I + E_{12}$ with respect to some basis.

Proof. By Proposition 2, the degrees n and m of the groups coincide. If $n = m = 2$ or $n = m = 3$, the assertion follows from Lemma 2.

Now suppose that $n = m \geq 4$. The proof of Proposition 2 shows that we can distinguish whether or not the fields K and L contain nontrivial cube roots of unity. If they do, the assertion follows from Lemma 4; if they do not and $n = m \geq 6$, it follows from Lemma 8. Thus two cases remain: the fields contain no nontrivial cube roots of unity, and $n = m = 4$ or $n = m = 5$.

First consider the case $n = m = 4$. Since the matrices $I + E_{12}$ and $I + E_{13}$ are conjugate, it suffices to prove that the image of $I + E_{13}$ has the

form $I + E_{13}$ with respect to some basis. Indeed, matrices of order 4 have one of the following forms with respect to some basis:

$$A_1 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ or } A_2 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Add the condition that the centraliser is noncommutative (i.e. add an assertion that two additional matrices exist; the formula remains existential). Only the matrix A_2 satisfies this condition. We have $A_2^2 = I + E_{13}$, which gives the required result.

Now let $n = 5$. As we already know, a matrix of the largest order that is a power of two is a Jordan block of maximal size with eigenvalue 1. Its fourth power equals $I + E_{15}$. Thus the image of $I + E_{15}$ under a finite partial isomorphism has the same form with respect to some basis, which proves the assertion of the lemma. $\square$

We now know that the form of $I + E_{12}$ is preserved under a special finite partial isomorphism. We can therefore use the result from the advanced course [6], namely Lemmas 2.20 and 2.21. This gives the following lemma.

Lemma 12. Let K and L be infinite fields of characteristic 2, and suppose that $G_n(K) \equiv_{\forall} G_n(L)$ ($G = \text{GL}, \text{SL}$). Then, under every special finite partial isomorphism

$$\Phi: G_n(K) \rightarrow G_n(L) \quad (G = \text{GL}, \text{SL})$$

all matrices of the form $I + E_{ij}$ map to matrices that, up to an automorphism of the group, have the same form with respect to a common basis.

2.2.4 Proof of the main theorem

Definition 10. For a given nonzero element α of the field K , we call $I + \alpha E_{12} \in \text{GL}_n(K)$ ($I + \alpha E_{12} \in \text{SL}_n(K)$) the *addition matrix*, and $\text{diag}[\alpha, 1/\alpha, 1, \dots, 1]$ the *multiplication matrix*.

Lemma 13. The product of the addition matrices corresponding to α and β is the addition matrix corresponding to $\alpha + \beta$. The product of the multiplication matrices corresponding to α and β is the multiplication matrix corresponding to $\alpha\beta$.

Proof. By direct verification. $\square$

Lemma 14. Let K and L be infinite fields of characteristic 2 such that

$$G_n(K) \equiv_{\forall} G_m(L) \quad (G = GL \text{ or } SL).$$

Then, under every special finite partial isomorphism

$$\Phi: G_n(K) \rightarrow G_m(L) \quad (G = GL, SL)$$

addition and multiplication matrices map to matrices of the same form. Moreover, if $\Phi(\text{diag}[\alpha, 1/\alpha, 1, \dots, 1]) = \text{diag}[\beta, 1/\beta, 1, \dots, 1]$ and $\Phi(I + \alpha E_{12}) = I + \gamma E_{12}$, then $\beta = \gamma$.

Proof. The centraliser of $I + E_{ij}$ ($i \neq j$) consists of all matrices (a_{st}) satisfying $a_{si} = 0$, $1 \leq s \leq n$, $s \neq i$, $a_{jt} = 0$, $1 \leq t \leq n$, $j \neq t$, and $a_{ii} = a_{jj}$. It follows that addition matrices are characterised by the following property: they are matrices of order 2 that commute with exactly those matrices among $I + E_{ij}$ ($i \neq j$) that commute with $I + E_{12}$.

We prove that multiplication matrices are precisely the matrices A for which the following relations hold:

$$\begin{aligned} A(I + E_{ij}) &= (I + E_{ij})A \quad (3 \leq i \leq n, 3 \leq j \leq n, i \neq j), \\ (A(I + E_{12})A^{-1})(I + E_{12}) &= (I + E_{12})(A(I + E_{12})A^{-1}), \\ (A(I + E_{21})A^{-1})(I + E_{21}) &= (I + E_{21})(A(I + E_{21})A^{-1}), \\ (\sigma_{12}A)^2 &= I, \\ (\sigma_{13}A\sigma_{13})(\sigma_{23}A\sigma_{23}) &= A, \text{ where } \sigma_{ij} = (I + E_{ij})(I + E_{ji})(I + E_{ij}). \end{aligned}$$

(If a relation does not apply because of the size of the matrices, we simply omit it.)

First consider the case $n = 2$ or $n \geq 4$. In this case, the first condition implies that

$$A = \begin{pmatrix} a_{11} & a_{12} & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ 0 & 0 & \lambda & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \lambda \end{pmatrix}.$$

Next,

$$\begin{aligned}
& A(I + E_{12})A^{-1} = \\
& = \begin{pmatrix} a_{11} & a_{12} & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ 0 & 0 & \lambda & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \lambda \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & 1 \end{pmatrix} \begin{pmatrix} \frac{a_{22}}{a_{11}a_{22}+a_{12}a_{21}} & \frac{a_{12}}{a_{11}a_{22}+a_{12}a_{21}} & 0 & \dots & 0 \\ \frac{a_{21}}{a_{11}a_{22}+a_{12}a_{21}} & \frac{a_{11}}{a_{11}a_{22}+a_{12}a_{21}} & 0 & \dots & 0 \\ 0 & 0 & \frac{1}{\lambda} & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \frac{1}{\lambda} \end{pmatrix} = \\
& = \begin{pmatrix} a_{11} & a_{12} & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ 0 & 0 & \lambda & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \lambda \end{pmatrix} \begin{pmatrix} \frac{a_{21}+a_{22}}{a_{11}a_{22}+a_{12}a_{21}} & \frac{a_{11}+a_{12}}{a_{11}a_{22}+a_{12}a_{21}} & 0 & \dots & 0 \\ \frac{a_{21}}{a_{11}a_{22}+a_{12}a_{21}} & \frac{a_{11}}{a_{11}a_{22}+a_{12}a_{21}} & 0 & \dots & 0 \\ 0 & 0 & \frac{1}{\lambda} & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \frac{1}{\lambda} \end{pmatrix} = \\
& = \begin{pmatrix} * & * & 0 & \dots & 0 \\ \frac{a_{21}^2}{a_{11}a_{22}+a_{12}a_{21}} & * & 0 & \dots & 0 \\ 0 & 0 & 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & 1 \end{pmatrix}
\end{aligned}$$

commutes with $I + E_{12}$, so $a_{21} = 0$. Similarly, $a_{12} = 0$.

Now apply the fourth relation. Since

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & I \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & I \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & I \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & I \end{pmatrix},$$

we obtain

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & I \end{pmatrix} \begin{pmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & \lambda I \end{pmatrix} = \begin{pmatrix} 0 & a_{22} & 0 \\ a_{11} & 0 & 0 \\ 0 & 0 & \lambda I \end{pmatrix}, \\
& \begin{pmatrix} 0 & a_{22} & 0 \\ a_{11} & 0 & 0 \\ 0 & 0 & \lambda I \end{pmatrix}^2 = \begin{pmatrix} a_{11}a_{22} & 0 & 0 \\ 0 & a_{11}a_{22} & 0 \\ 0 & 0 & \lambda^2 I \end{pmatrix} = I,
\end{aligned}$$

and hence $a_{11}a_{22} = 1$, $\lambda = 1$.

Consider the case $n = 3$. Here we omit the first condition. The second condition says that the matrix $A(I + E_{12})A^{-1} = I + AE_{12}A^{-1}$ commutes with $I + E_{12}$, so $AE_{12}A^{-1}$ commutes with E_{12} . Write $A = (a_{ij})_{i,j=1}^3$, $A^{-1} =$

$(b_{ij})_{i,j=1}^3$. In this notation,

$$AE_{12}A^{-1} = \begin{pmatrix} a_{11}b_{21} & a_{11}b_{22} & a_{11}b_{23} \\ a_{21}b_{21} & a_{21}b_{22} & a_{21}b_{23} \\ a_{31}b_{21} & a_{31}b_{22} & a_{31}b_{23} \end{pmatrix},$$

and the condition of commutation with E_{12} takes the form:

$$\begin{pmatrix} 0 & a_{11}b_{21} & 0 \\ 0 & a_{21}b_{21} & 0 \\ 0 & a_{31}b_{21} & 0 \end{pmatrix} = \begin{pmatrix} a_{21}b_{21} & a_{21}b_{22} & a_{21}b_{23} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This relation gives the following conditions on the entries:

$$a_{21}b_{21} = 0, \quad a_{11}b_{21} = a_{21}b_{22}, \quad a_{21}b_{23} = 0, \quad a_{31}b_{21} = 0.$$

First we prove that $a_{21} = 0$. Otherwise, b_{21}, b_{22}, b_{23} must all be zero, contrary to the invertibility of A^{-1} .

Next, the conditions $a_{11}b_{21} = a_{21}b_{22} = 0$ and $a_{31}b_{21} = 0$, together with the fact that at least one of the entries a_{11}, a_{31} must be nonzero, imply $b_{21} = 0$.

Express b_{21} in terms of the entries of A :

$$b_{21} = -\frac{\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}}{\det A} = \frac{a_{23}a_{31} - a_{21}a_{33}}{\det A} = \frac{a_{23}a_{31}}{\det A} = 0,$$

hence $a_{23}a_{31} = 0$.

Now that we know that A satisfies $a_{21} = 0$ and $a_{23}a_{31} = 0$, apply the last relation. Compute the factors on the left:

$$\sigma_{13}A\sigma_{13} = \begin{pmatrix} a_{33} & a_{32} & a_{31} \\ a_{23} & a_{22} & 0 \\ a_{13} & a_{12} & a_{11} \end{pmatrix}, \quad \sigma_{23}A\sigma_{23} = \begin{pmatrix} a_{11} & a_{13} & a_{12} \\ a_{31} & a_{33} & a_{32} \\ 0 & a_{23} & a_{22} \end{pmatrix}.$$

Computing the entry in position $(2, 1)$ of the product gives:

$$a_{23}a_{11} + a_{22}a_{31} = 0.$$

Consider two cases: 1) $a_{23} = 0$ and 2) $a_{31} = 0$.

1) If $a_{23} = 0$, then in the invertible matrix $\sigma_{13}A\sigma_{13}$, the second-row entry a_{22} must be nonzero. Thus $a_{22}a_{31} = 0$ implies $a_{31} = 0$.

2) If $a_{31} = 0$, then in the invertible matrix $\sigma_{23}A\sigma_{23}$, the first-column entry a_{11} must be nonzero. Thus $a_{23}a_{11} = 0$ implies $a_{23} = 0$.

Thus, in either case, $a_{23} = 0$ and $a_{31} = 0$.

Consider a few more entries of the product $(\sigma_{13}A\sigma_{13})(\sigma_{23}A\sigma_{23}) = A$:

$$\begin{pmatrix} a_{33} & a_{32} & 0 \\ 0 & a_{22} & 0 \\ a_{13} & a_{12} & a_{11} \end{pmatrix} \cdot \begin{pmatrix} a_{11} & a_{13} & a_{12} \\ 0 & a_{33} & a_{32} \\ 0 & 0 & a_{22} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & 0 \\ 0 & a_{32} & a_{33} \end{pmatrix}.$$

The entry in position $(1, 1)$ gives $a_{33}a_{11} = a_{11}$ ($a_{11} \neq 0$), so $a_{33} = 1$.

In position $(3, 1)$, we have $a_{13}a_{11} = 0$, hence $a_{13} = 0$.

In position $(3, 2)$, we have $a_{13}^2 + a_{12}a_{33} = 0 + a_{12} = a_{32}$, hence $a_{12} = a_{32}$.

In position $(1, 3)$, we have $a_{33}a_{12} + a_{32}^2 = a_{13}$, hence $a_{12} + a_{12}^2 = 0$, so $a_{12} = 0$ or $a_{12} = 1$. If we assume that $a_{12} = a_{32} = 1$, then the entry in position $(3, 3)$ gives $a_{13}a_{12} + a_{12}a_{32} + a_{11}a_{22} = a_{33}$, that is, $1 + a_{11}a_{22} = 1$, and hence $a_{11}a_{22} = 0$, contrary to the invertibility of A . Thus the only possibility is $a_{12} = a_{32} = 0$ and $a_{11}a_{22} = 1$. The matrix A therefore has the required form.

We now prove the second assertion of the lemma. Suppose that $\Phi(I + \frac{1}{\alpha}E_{21}) = I + \delta E_{21}$ (preservation of this form is proved in the same way as for $I + \alpha E_{12}$). The following relation holds in $\text{GL}_n(K)$ ($\text{SL}_n(K)$):

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \frac{1}{\alpha} & 1 \end{pmatrix} \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ 0 & \frac{1}{\alpha} \end{pmatrix},$$

so in $\text{GL}_n(L)$ ($\text{SL}_n(L)$) we have:

$$\begin{aligned} \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \delta & 1 \end{pmatrix} \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \\ = \begin{pmatrix} \gamma^2\delta & 1 + \gamma\delta \\ 1 + \gamma\delta & \delta \end{pmatrix} = \begin{pmatrix} \beta & 0 \\ 0 & \frac{1}{\beta} \end{pmatrix}. \end{aligned}$$

It follows that $\gamma\delta = 1$ and $\beta = \gamma^2\delta = \gamma$, as required. $\square$

Proof of Theorem 6. The equality of the degrees of the groups was established in Proposition 2. We use the criterion for universal equivalence to show that the fields K and L are universally equivalent.

Let $K_1 \subset K$ be a finite partial structure of the field K . Our task is to find a partial structure L_1 of the field L partially isomorphic to it. Let $G_1 \subset \text{G}_n(K)$ ($\text{G} = \text{GL}, \text{SL}$) contain all addition matrices $I + \alpha E_{12}$ and multiplication matrices $\text{diag}[\alpha, \frac{1}{\alpha}, 1, \dots, 1]$ for all $\alpha \in K_1^*$, a set of pairwise commuting matrices of order 3 forming the system $\mathcal{A}$ or $\mathcal{B}$ depending on whether the fields contain nontrivial cube roots of unity, the identity matrix, all matrices $I + E_{ij}$ ($i, j = 1, \dots, n; i \neq j$), and finitely many auxiliary matrices from Lemmas 6, 10, and Proposition 2. Since the groups are universally

equivalent, for the partial structure G_1 there exists a finite partial structure $G_2 \subset G_n(L)$ ($G = \text{GL}, \text{SL}$) partially isomorphic to G_1 , with the finite partial isomorphism $\Phi: G_1 \rightarrow G_2$ being special. Thus addition and multiplication matrices map to matrices of the same form. If an addition matrix S and a multiplication matrix P correspond to the same element α of the field K , their images $\Phi(S)$ and $\Phi(P)$ correspond to the same element β of the field L . Then L_1 is the set of all elements of L corresponding to the pairs of images of the addition and multiplication matrices for all elements of K_1 . This proves the theorem. $\square$

2.3 Fields of characteristic different from 2

In this section, we prove the following theorem.

Theorem 7. Let K and L be infinite fields, $\text{char } K, \text{char } L \neq 2$, and suppose that the groups $G_n(K)$ and $G_m(L)$ ($G = \text{GL}, \text{SL}$, $n, m \geq 2$) are universally equivalent. Then $n = m$ and the fields K and L are universally equivalent.

Throughout this section, all fields considered have characteristic different from 2, unless explicitly stated otherwise.

2.3.1 Distinguishing characteristic 2 by existential formulas

Lemma 15. Let K be a field, $\text{char } K \neq 2$, and let n be a positive integer. The group $\text{GL}_n(K)$ has a set of $2^n - 1$ pairwise commuting matrices of order 2 and no larger such set; likewise, $\text{SL}_n(K)$ has a set of $2^{n-1} - 1$ pairwise commuting matrices of order 2 and no larger such set.

Proof. There is a basis with respect to which all the pairwise commuting matrices of order 2 are diagonal and have the form $\text{diag}[\pm 1, \dots, \pm 1]$. Thus, $\text{GL}_n(K)$ contains $2^n - 1$ pairwise commuting matrices of order 2, whereas $\text{SL}_n(K)$ contains $2^{n-1} - 1$. $\square$

Let $\mathcal{D}(\text{GL}_n(K))$ ($\mathcal{D}(\text{SL}_n(K))$), or simply $\mathcal{D}$ when the group is clear from the context) denote a fixed maximal set of pairwise commuting matrices of order 2, whose existence was proved in Lemma 15. Clearly, the set of matrices $\mathcal{D}$ can be defined by an existential formula.

Note that Lemma 15 allows us to distinguish between fields of characteristic 2 and fields of characteristic different from 2. Indeed, we have the following lemma.

Lemma 16. Let K and L be infinite fields of arbitrary characteristic, and suppose that the groups $G_n(K)$ and $G_m(L)$ ($G = GL, SL$) are universally equivalent. Then either $\text{char } K = \text{char } L = 2$, or $\text{char } K \neq 2$ and $\text{char } L \neq 2$.

Proof. Consider the following formula:

$$\begin{aligned} \exists A_1 \dots \exists A_M \left(\bigwedge_{i=1}^M (\neg A_i = I) \& \bigwedge_{\substack{i,j=1 \\ i \neq j}}^M (\neg A_i = A_j) \& \right. \\ \left. \& \bigwedge_{i,j=1}^M (A_i A_j = A_j A_i) \& \bigwedge_{i=1}^M (A_i^2 = I) \right). \end{aligned}$$

If this formula is false in both groups for some sufficiently large M , then both fields have characteristic different from two. Otherwise, the fields have characteristic two. Indeed, since the field is infinite, for any arbitrarily large M we can take $A_1, \dots, A_M$ to be matrices of the form $I + \alpha_i E_{12}$, where $\alpha_1, \dots, \alpha_M$ are distinct nonzero elements of the field. $\square$

2.3.2 Determining the dimension

The pairwise commuting involutions in the system $\mathcal{D}$ split into conjugacy classes, with each class consisting of matrices that have the same number of (-1) entries on the diagonal.

Proposition 3. Let K and L be infinite fields of characteristic different from 2, and suppose that $G_n(K) \equiv_{\forall} G_m(L)$ ($G = GL, SL$). Then $n = m$.

Proof. Consider the formula:

$$\begin{aligned} \exists A_1 \dots \exists A_{2^N-1} \left(\bigwedge_{i=1}^{2^N-1} (\neg A_i = I) \& \bigwedge_{\substack{i,j=1 \\ i \neq j}}^{2^N-1} (\neg A_i = A_j) \& \right. \\ \left. \& \bigwedge_{i,j=1}^{2^N-1} (A_i A_j = A_j A_i) \& \bigwedge_{i=1}^{2^N-1} (A_i^2 = I) \right). \quad (2.4) \end{aligned}$$

Since the groups are universally equivalent, the largest N for which formula (2.4) holds in the groups is the same for both groups. For general linear groups, $n = m = N$, and for special linear groups, $n = m = N + 1$. $\square$

2.3.3 Properties of involutions

Lemma 17. Let K be an infinite field, $\text{char } K \neq 2$. In the group $\text{GL}_n(K)$, the two subsets of the system $\mathcal{D}$ consisting of matrices in which (-1) occurs once or $(n-1)$ times on the diagonal can be defined by an existential formula. If n is odd, then in the group $\text{SL}_n(K)$, the subset of the system $\mathcal{D}$ consisting of matrices in which (-1) occurs $(n-1)$ times on the diagonal can be defined by an existential formula.

Proof. The assertion follows from the fact that, among all involutions, those we need lie in $\text{GL}_n(K)$ in one (for $n = 2$) or two (for $n > 2$) conjugacy classes, each containing n elements, and in $\text{SL}_n(K)$ in a single conjugacy class of n elements; all other conjugacy classes (except the class consisting of the single matrix $-I$ in the case of $\text{GL}_n(K)$) contain more involutions. $\square$

Definition 11. Let $\mathcal{E}$ denote the subset of matrices in the system $\mathcal{D}$ in which (-1) occurs once or $(n-1)$ times on the diagonal (if both possibilities occur, choose either of these two subsets as $\mathcal{E}$). By the previous lemma, this set of matrices can be defined by an existential formula.

Lemma 18. Let K and L be infinite fields of characteristic different from 2, and suppose that $\text{G}_n(K) \equiv_{\forall} \text{G}_n(L)$ ($\text{G} = \text{GL}, \text{SL}$), with n odd in the case of special linear groups. Then under any finite partial isomorphism

$$\Phi: \text{G}_n(K) \rightarrow \text{G}_n(L) \text{ (G = GL, SL)}$$

such that $\text{dom } \Phi$ contains all matrices in the system $\mathcal{D}$, a diagonal matrix is mapped to a diagonal matrix.

Proof. The assertion follows from the fact that a diagonal matrix commutes with all matrices in the system $\mathcal{E}$. Its image therefore has the same property and is consequently diagonal. $\square$

Lemma 19. Let K and L be infinite fields of characteristic different from 2, and suppose that $\text{G}_n(K) \equiv_{\forall} \text{G}_n(L)$ ($\text{G} = \text{GL}, \text{SL}, n$ odd for special linear groups). Then under any finite partial isomorphism

$$\Phi: \text{G}_n(K) \rightarrow \text{G}_n(L), \text{G} = \text{GL}, \text{SL},$$

such that $\text{dom } \Phi$ contains all matrices in the system $\mathcal{D}$ and a finite set of auxiliary matrices from the previous lemmas, the matrix $-I + E_{11} + E_{22} + E_{12} + E_{21}$ is mapped to a matrix of the form

$$\begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & \pm 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \pm 1 \end{pmatrix}$$

with respect to a basis relative to which the form of the matrices in the system $\mathcal{D}$ remains unchanged.

Proof. As we already know, the set of matrices $\mathcal{E}$ can be defined by an existential formula. The matrix $-I + E_{11} + E_{22} + E_{21} + E_{12}$ commutes with all but two matrices in the system $\mathcal{E}$. Therefore,

$$\Phi(-I + E_{11} + E_{22} + E_{21} + E_{12}) = \begin{pmatrix} d_{11} & d_{12} & 0 & \dots & 0 \\ d_{21} & d_{22} & 0 & \dots & 0 \\ 0 & 0 & d_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_{nn} \end{pmatrix},$$

and since $(-I + E_{11} + E_{22} + E_{21} + E_{12})^2 = I$, we have

$$\begin{aligned} I &= \begin{pmatrix} d_{11} & d_{12} & 0 & \dots & 0 \\ d_{21} & d_{22} & 0 & \dots & 0 \\ 0 & 0 & d_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_{nn} \end{pmatrix}^2 = \\ &= \begin{pmatrix} d_{11}^2 + d_{12}d_{21} & d_{11}d_{12} + d_{12}d_{22} & 0 & \dots & 0 \\ d_{21}d_{11} + d_{22}d_{21} & d_{21}d_{12} + d_{22}^2 & 0 & \dots & 0 \\ 0 & 0 & d_{33}^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_{nn}^2 \end{pmatrix}. \end{aligned}$$

Further, $(-I + E_{11} + E_{22} + E_{21} + E_{12})(-I + 2E_{11})(-I + E_{11} + E_{22} + E_{21} + E_{12}) = -I + 2E_{22}$ (and similarly when the matrices in the system $\mathcal{E}$ have one occurrence of (-1) on the diagonal). Hence the following relation holds (we write it only for the upper-left block):

$$\begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

and consequently,

$$\begin{pmatrix} d_{11}^2 - d_{12}d_{21} & d_{11}d_{12} - d_{12}d_{22} \\ d_{11}d_{21} - d_{21}d_{22} & d_{12}d_{21} - d_{22}^2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The relations above give $d_{12}d_{22} = 0$. If we assume that $d_{12} = 0$, then $d_{22}^2 = 1$ and $-d_{22}^2 = 1$, a contradiction. Hence $d_{22} = 0$. Next, $d_{12}d_{21} + d_{22}^2 = 1$

gives $d_{12}d_{21} = 1$, and therefore $d_{11} = 0$. Thus, the image of the matrix $-I + E_{11} + E_{22} + E_{21} + E_{12}$ has the form:

$$A_\alpha = \begin{pmatrix} 0 & \alpha & 0 & \dots & 0 \\ \frac{1}{\alpha} & 0 & 0 & \dots & 0 \\ 0 & 0 & \pm 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \pm 1 \end{pmatrix}.$$

Changing the basis using the matrix $\text{diag}[1/\alpha, 1, \dots, 1]$ gives the required result. $\square$

2.3.4 Defining transvections in $\text{GL}_n(K)$ and, for odd n , in $\text{SL}_n(K)$

Lemma 20. Let K, L be infinite fields, $\text{char } K, \text{char } L \neq 2$. Under any finite partial isomorphism between the groups $\text{G}_n(K)$ and $\text{G}_n(L)$ ($\text{G} = \text{GL}, \text{SL}$ and n odd in the case of special linear groups) whose domain contains all pairwise commuting involutions, the matrix $-I + E_{11} + E_{22} + E_{12} + E_{21}$, the matrix $\text{diag}[2, 1, \frac{1}{2}, 1, \dots, 1]$ (the matrix $\text{diag}[2, 1]$ when $\text{G} = \text{GL}$ and $n = 2$), and a finite set of auxiliary matrices, a set of matrices of the form $I + \alpha_i E_{12}, i = 1, \dots, k$ is mapped to matrices of the same form.

Proof. We already know that the set of pairwise commuting involutions retains its form under a finite partial isomorphism, the matrix $-I + E_{11} + E_{22} + E_{12} + E_{21}$ is mapped to $E_{12} + E_{21} + \text{diag}[0, 0, \pm 1, \dots, \pm 1]$, and diagonal matrices are mapped to diagonal matrices.

We now consider the matrix $I + \alpha E_{12}$. It commutes with exactly those matrices in the set $\mathcal{E}$ that commute with $-I + E_{11} + E_{22} + E_{12} + E_{21}$, so its image has the form

$$\begin{pmatrix} a & b & 0 & \dots & 0 \\ c & d & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{pmatrix}.$$

Consider two cases: $\text{char } K = 3$ and $\text{char } K \neq 3$.

If $\text{char } K = 3$, then $\text{char } L = 3$; otherwise, $\text{G}_n(\overline{L})$ contains at most 3^n distinct pairwise commuting matrices of order 3, whereas $\text{G}_n(K)$ contains arbitrarily many such matrices: $I + \beta_i E_{12}$, where β_i ranges over all elements of the field.

First, consider the case $n = 2$ (for general linear groups only). In characteristic 3, the matrix $I + \alpha E_{12}$ satisfies the relation $(I + \alpha E_{12})^{-1} = I - \alpha E_{12} = (-E_{11} + E_{22})(I + \alpha E_{12})(-E_{11} + E_{22})$, and the image of the matrix $-E_{11} + E_{22}$ is an involution in the system $\mathcal{E}$. Therefore, the image of $I + \alpha E_{12}$ satisfies

$$\begin{pmatrix} a & -b \\ -c & d \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Now let $n > 2$. Choose two matrices in the system $\mathcal{E}$: one that commutes with $I + \alpha E_{12}$ and one that does not. Their product is a matrix of the form $\text{diag}[\pm 1, \dots, \pm 1]$ in which (-1) occurs exactly 2 times on the diagonal; we may assume that it is the matrix $I - 2E_{11} - 2E_{33}$. Since the relations $(I - 2E_{11} - 2E_{33})(I + \alpha E_{12})(I - 2E_{11} - 2E_{33}) = (I + \alpha E_{12})^{-1} = (I + \alpha E_{12})^2$ hold, we can write the following for the images (the relations in the upper-left block are the same as for $n = 2$):

$$\begin{aligned} \begin{pmatrix} a & -b & 0 & \dots & 0 \\ -c & d & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \lambda_n \end{pmatrix} &= \begin{pmatrix} d/(ad - bc) & -b/(ad - bc) & 0 & \dots & 0 \\ -c/(ad - bc) & a/(ad - bc) & 0 & \dots & 0 \\ 0 & 0 & 1/\lambda_3 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & 1/\lambda_n \end{pmatrix} = \\ &= \begin{pmatrix} a^2 + bc & (a + d)b & 0 & \dots & 0 \\ (a + d)c & bc + d^2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3^2 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \lambda_n^2 \end{pmatrix}. \end{aligned}$$

This equality gives $\lambda_3 = \dots = \lambda_n = 1$. Since b and c cannot both equal 0 (otherwise, the image of $I + \alpha E_{12}$ would commute with all matrices in the system $\mathcal{E}$, which is not the case), either $-b = -\frac{b}{ad - bc}$ or $-c = -\frac{c}{ad - bc}$ gives $ad - bc = 1$. We then observe that $a = d, a + d = -1$, so $a = d = 1$ and, finally, $bc = 0$. Since the images of the matrices $I + \alpha_i E_{12}$ commute pairwise, their nonzero off-diagonal entries occupy the same position with respect to a common basis. This proves the assertion when $\text{char } K = 3$.

Now let $\text{char } K \neq 3$; then $\text{char } L \neq 3$ as well. The following relation holds:

$$(\text{diag}[2, 1, \frac{1}{2}, 1, \dots, 1])(I + \alpha E_{12})(\text{diag}[2, 1, \frac{1}{2}, 1, \dots, 1])^{-1} = (I + \alpha E_{12})^2.$$

The image of $\text{diag}[2, 1, \frac{1}{2}, 1, \dots, 1]$ under a finite partial isomorphism is a diagonal matrix $\text{diag}[\alpha_1, \dots, \alpha_n]$. Moreover, since the matrix $(\text{diag}[2, 1, \frac{1}{2}, 1, \dots, 1])^2$ does not commute with $-I + E_{11} + E_{12} + E_{21} + E_{22}$, the matrix $(\text{diag}[\alpha_1, \dots, \alpha_n])^2$ does not commute with $-I + E_{11} + E_{12} + E_{21} + E_{22}$ either, so $\alpha_1^2 \neq \alpha_2^2$. We can now write the relation for the images:

$$\begin{pmatrix} a & \frac{b\alpha_1}{\alpha_2} & 0 & \dots & 0 \\ \frac{c\alpha_2}{\alpha_1} & d & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{pmatrix} = \begin{pmatrix} \alpha_1 & 0 & \dots & \dots & 0 \\ 0 & \ddots & & & \vdots \\ \vdots & & \ddots & & \vdots \\ \vdots & & & \ddots & 0 \\ 0 & \dots & \dots & 0 & \alpha_n \end{pmatrix} \begin{pmatrix} a & b & 0 & \dots & 0 \\ c & d & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{pmatrix}.$$

$$\cdot \begin{pmatrix} \frac{1}{\alpha_1} & 0 & \dots & \dots & 0 \\ 0 & \ddots & & & \vdots \\ \vdots & & \ddots & & \vdots \\ \vdots & & & \ddots & 0 \\ 0 & \dots & \dots & 0 & \frac{1}{\alpha_n} \end{pmatrix} = \begin{pmatrix} a^2 + bc & ab + bd & 0 & \dots & 0 \\ ac + cd & bc + d^2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n^2 \end{pmatrix}.$$

Hence $\lambda_i = 1, i = 3, \dots, n$. If $bc \neq 0$, then, since $\frac{b\alpha_1}{\alpha_2} = b(a + d)$ and $\frac{c\alpha_2}{\alpha_1} = c(a + d)$, we have $\frac{\alpha_1}{\alpha_2} = a + d = \frac{\alpha_2}{\alpha_1}$ and therefore $\alpha_1^2 = \alpha_2^2$, contradicting what was proved above. Thus, $bc = 0$ and $a = d = 1$. $\square$

Lemma 21. Let K, L be infinite fields, $\text{char } K, \text{char } L \neq 2$. Under any finite partial isomorphism between the groups $G_n(K)$ and $G_n(L)$ ($G = \text{GL}, \text{SL}$, with n odd in the case of special linear groups) whose domain contains all pairwise commuting involutions, the matrix $-I + E_{11} + E_{12} + E_{21} + E_{22}$, the matrix $\text{diag}[2, 1, 1/2, 1, \dots, 1]$ ($\text{diag}[2, 1]$ when $G = \text{GL}, n = 2$), and a finite set of auxiliary matrices from the previous lemmas, the matrix $I + E_{12}$ is mapped to $I + E_{12}$.

Proof. We have already proved that the image of $I + E_{12}$ is a matrix $I + \alpha E_{12}$. Let us show that $\alpha = 1$. We write the relations satisfied by elements in the

domain of the finite partial isomorphism (showing only the upper-left block):

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \\ \left[\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \left[\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right],$$

Thus, the matrix $I + \alpha E_{12}$ satisfies the relations:

$$\begin{pmatrix} 0 & 1 \\ 1 & \alpha \end{pmatrix} = \left[\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \left[\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \\ = \begin{pmatrix} \alpha & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \alpha^2 - 1 & \alpha \\ \alpha & 1 \end{pmatrix},$$

which immediately give $\alpha = 1$. □

2.3.5 Preservation of addition and multiplication matrices in $\text{GL}_n(K)$ and, for odd n , in $\text{SL}_n(K)$

Lemma 22. Let K, L be infinite fields, $\text{char } K, \text{char } L \neq 2$. Under any finite partial isomorphism between the groups $\text{G}_n(K)$ and $\text{G}_n(L)$ ($\text{G} = \text{GL}, \text{SL}$, with n odd when $\text{G} = \text{SL}$) whose domain contains the finite set of matrices used to prove the previous lemmas, addition matrices are mapped to addition matrices, and multiplication matrices are mapped to multiplication matrices. Moreover, if two matrices correspond to the same element of the field K , their images correspond to the same element of the field L .

Proof. We have already proved that the form of addition matrices is preserved under a finite partial isomorphism. A multiplication matrix can be expressed as follows (we write the relation only for the upper-left block):

$$\begin{pmatrix} \alpha & 0 \\ 0 & \frac{1}{\alpha} \end{pmatrix} = \begin{pmatrix} 1 & -\alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \frac{1}{\alpha} & 1 \end{pmatrix} \begin{pmatrix} 1 & -\alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \\ = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}^{-1} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{\alpha} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}^{-1} \cdot \\ \cdot \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

Then the upper-left block of the image of the multiplication matrix under consideration (a diagonal matrix whose diagonal entries, except for the first

two, are all one) is equal to the product:

$$\begin{aligned}
& \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}^{-1} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}^{-1} \\
& \quad \cdot \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \\
& = \begin{pmatrix} 1 & -\beta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix} \begin{pmatrix} 1 & -\beta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \\
& \qquad \qquad \qquad = \begin{pmatrix} 2\beta - \beta^2\gamma & 1 - \beta\gamma \\ \beta\gamma - 1 & \gamma \end{pmatrix},
\end{aligned}$$

Consequently, $\beta\gamma = 1$, and the image of the multiplication matrix under consideration is $I + (\beta - 1)E_{11} + \left(\frac{1}{\beta} - 1\right)E_{22}$, as required. $\square$

2.3.6 The case of $\text{SL}_2(K)$

Lemma 23. Let K be an infinite field, $\text{char } K \neq 2$. Then a matrix in $\text{SL}_2(K)$ that has the form $I + E_{12}$ with respect to some basis can be defined by an existential formula.

Proof. We prove that a matrix A satisfying the formula

$$\begin{aligned}
\exists X \exists Y ((A^2 \neq I) \& (XAX^{-1}A = AXAX^{-1}) \& \\
& \& (X^2A \neq AX^2) \& (YAY^{-1} = A^4)),
\end{aligned}$$

has the required form with respect to some basis.

Note that the formula holds for the matrix $A = I + E_{12}$ when $X = \alpha E_{11} + \frac{1}{\alpha} E_{22}$, $\alpha \in K^*$, $\alpha^4 \neq 1$, $Y = 2E_{11} + \frac{1}{2}E_{22}$.

Now suppose that the formula above holds for a matrix A . If its Jordan form over $\overline{K}$ is not diagonal, then, since $A \in \text{SL}_2(K)$, it is equal to

$$A_+ = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ or } A_- = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

In the first case, the proof is complete. The second case is impossible because the matrix $A_-^4 = I - 4E_{12}$ is not conjugate to A_- (the former has eigenvalue 1, whereas the latter has eigenvalue (-1)).

If the matrix A is diagonalisable over $\overline{K}$, then its diagonal form is $\alpha E_{11} + \frac{1}{\alpha} E_{22}$, where $\alpha \neq \pm 1$ because $A^2 \neq I$. Write $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and consider the

relations in our formula:

$$\begin{aligned}
XAX^{-1}A &= \begin{pmatrix} \alpha a & \frac{b}{\alpha} \\ c\alpha & \frac{d}{\alpha} \end{pmatrix} \begin{pmatrix} \alpha d & -\frac{b}{\alpha} \\ -c\alpha & \frac{a}{\alpha} \end{pmatrix} = \begin{pmatrix} ad\alpha^2 - bc & -ab + \frac{ab}{\alpha^2} \\ cd\alpha^2 - cd & -bc + \frac{ad}{\alpha^2} \end{pmatrix} = \\
&= \begin{pmatrix} ad\alpha^2 - bc & ab - ab\alpha^2 \\ -\frac{cd}{\alpha^2} + cd & -bc + \frac{ad}{\alpha^2} \end{pmatrix} = \begin{pmatrix} \alpha a & \alpha b \\ \frac{c}{\alpha} & \frac{d}{\alpha} \end{pmatrix} \begin{pmatrix} \alpha d & -\alpha b \\ -\frac{c}{\alpha} & \frac{a}{\alpha} \end{pmatrix} = AXAX^{-1}, \\
X^2A &= \begin{pmatrix} (a^2 + bc)\alpha & \frac{ab+bd}{\alpha} \\ (ac + cd)\alpha & \frac{bc+d^2}{\alpha} \end{pmatrix} \neq \begin{pmatrix} (a^2 + bc)\alpha & (ab + bd)\alpha \\ \frac{ac+cd}{\alpha} & \frac{bc+d^2}{\alpha} \end{pmatrix} = AX^2.
\end{aligned}$$

The condition $XAX^{-1}A = AXAX^{-1}$ gives

$$cd \left(2 - \alpha^2 - \frac{1}{\alpha^2} \right) = 0 \text{ and } ab \left(2 - \alpha^2 - \frac{1}{\alpha^2} \right) = 0,$$

and since $\alpha \neq \pm 1$, we have $ab = cd = 0$. If we assume that $a = 0$, then $c \neq 0$ because X is nonsingular, so $d = 0$. Similarly, if $b = 0$, then $c = 0$. In both cases, we obtain

$$\frac{ab + bd}{\alpha} = 0 = (ab + bd)\alpha \text{ and } (ac + cd)\alpha = 0 = \frac{ac + cd}{\alpha},$$

which contradicts the condition $X^2A \neq AX^2$. Thus, A is not diagonalisable over $\overline{K}$, as required. $\square$

Lemma 24. Let K be an infinite field, $\text{char } K \neq 2$. Then a matrix in $\text{SL}_2(K)$ that has the form $-E_{12} + E_{21}$ with respect to some basis can be defined by an existential formula.

Proof. We already know how to define the matrix $I + E_{12}$ by an existential formula. Note that the following relations hold:

$$\begin{aligned}
\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \\
\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \\
\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{-1} &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.
\end{aligned}$$

Now let $B \in \text{SL}_2(K)$ be a matrix of order 4 such that

$$(I + E_{12})B(I + E_{12})B(I + E_{12}) = B \text{ and}$$

$$(I + E_{12})B^{-1}(I + E_{12})B^{-1}(I + E_{12}) = B.$$

Let $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, The relations above give

$$\begin{aligned}
& \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \\
& = \begin{pmatrix} a^2 + c(2a + b + c + d) & a(a + c) + b(a + c) + (c + d)(a + b + c + d) \\ ac + c^2 + cd & c(a + b + c + 2d) + d^2 \end{pmatrix} = \\
& = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \\
& = \begin{pmatrix} * & * \\ -ac + c^2 - cd & * \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.
\end{aligned}$$

Consequently, $c(a + d) = 0$. If we assume that $c = 0$, then $a^2 = a$ and $d^2 = d$, so $a = d = 1$; we then have $1 + b + 1 + b + 1 = b$ and $b = -3$. Since $\text{char } K \neq 2$,

$$\begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}^4 = \begin{pmatrix} 1 & -12 \\ 0 & 1 \end{pmatrix}$$

is the identity matrix only when $\text{char } K = 3$, but then B itself is the identity matrix. This is a contradiction. Hence $c \neq 0$.

Thus, $a = -d$. In this case, the equality $ac + c^2 + cd = c$ gives $c = 1$. If $a = 0$, then, since $B \in \text{SL}_2(K)$, we have $b = -1$, and the assertion is proved. Otherwise, change the basis using the matrix $I + aE_{12}$ (which commutes with $I + E_{12}$); this brings B to the form

$$\begin{pmatrix} 0 & b + a^2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

as required. $\square$

Lemma 25. Let K be an infinite field, $\text{char } K \neq 2$. Then a matrix in $\text{SL}_2(K)$ that has the form $\text{diag}[\alpha, \frac{1}{\alpha}]$ with respect to a basis relative to which the form of the matrices $I + E_{12}$ and $-E_{12} + E_{21}$ remains unchanged can be defined by an existential formula.

Proof. Note that a diagonal matrix has the following properties: $(\text{diag}[\alpha, \frac{1}{\alpha}])(I + E_{12})(\text{diag}[\alpha, \frac{1}{\alpha}])^{-1}$ commutes with $I + E_{12}$, and $(-E_{12} + E_{21})(\text{diag}[\alpha, \frac{1}{\alpha}]) = (\text{diag}[\alpha, \frac{1}{\alpha}])^{-1}(-E_{12} + E_{21})$.

Suppose that a matrix $aE_{11} + bE_{12} + cE_{21} + dE_{22}$ satisfies the properties of a diagonal matrix listed above. Then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - ac - bc & a^2 \\ -c^2 & ac - bc + ad \end{pmatrix}$$

commutes with $I + E_{12}$, so $c = 0$, $ad = 1$. Next,

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & \frac{1}{a} \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{a} \\ a & b \end{pmatrix} = \begin{pmatrix} -b & -\frac{1}{a} \\ a & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{a} & -b \\ 0 & a \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

so $b = 0$, and the matrix under consideration is diagonal. $\square$

Lemma 26. Let K, L be infinite fields, $\text{char } K, \text{char } L \neq 2$. Under any finite partial isomorphism between the groups $\text{SL}_2(K)$ and $\text{SL}_2(L)$ whose domain contains the set of matrices used in the proofs of the previous lemmas, addition matrices are mapped to addition matrices, and multiplication matrices are mapped to multiplication matrices. Moreover, if two matrices correspond to the same element of the field K , their images correspond to the same element of the field L .

Proof. The previous lemma shows that the form of multiplication matrices is preserved. Let us show that the form of addition matrices is also preserved. Let $I + \alpha E_{12}$ be an addition matrix. It commutes with $I + E_{12}$, so its image has the form $\pm I + \beta E_{12}$. Since

$$\begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}^{-1} = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}^4,$$

the image of $I + \alpha E_{12}$ is $I + \beta E_{12}$; otherwise, a diagonal matrix $\text{diag}[\gamma, \frac{1}{\gamma}]$, the image of $\text{diag}[2, \frac{1}{2}]$, would have to satisfy the relation:

$$\begin{pmatrix} \gamma & 0 \\ 0 & \frac{1}{\gamma} \end{pmatrix} \begin{pmatrix} -1 & \beta \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{1}{\gamma} & 0 \\ 0 & \gamma \end{pmatrix} = \begin{pmatrix} -1 & \beta \\ 0 & -1 \end{pmatrix}^4,$$

but the left-hand side is equal to

$$\begin{pmatrix} -1 & \beta\gamma^2 \\ 0 & -1 \end{pmatrix},$$

and the right-hand side is

$$\begin{pmatrix} 1 & -4\beta \\ 0 & 1 \end{pmatrix},$$

Thus, we have obtained a contradiction.

It remains to prove that the images of an addition matrix and a multiplication matrix corresponding to the same element of the field K correspond to the same element of the field L . The proof is analogous to that of Lemma 22, once we note that

$$\begin{pmatrix} 1 & 0 \\ \frac{1}{\alpha} & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 1 & -\frac{1}{\alpha} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

$\square$

2.3.7 The case of $\text{SL}_n(K)$ for even $n \geq 4$

Lemma 27. Let K be an infinite field, $\text{char } K \neq 2$, $n \geq 4$, with n even. Then, in the group $\text{SL}_n(K)$, the subset of the system $\mathcal{D}$ consisting of matrices with exactly two (-1) entries on the diagonal can be defined by an existential formula.

Proof. The set $\mathcal{D}$ splits into conjugacy classes. The subsets consisting of matrices with $(n-2)$ or two (-1) entries on the diagonal are two conjugacy classes, each containing C_n^2 elements. All other conjugacy classes, except the class consisting of the single element $-I$, contain more elements. If $n = 4$, there is only one conjugacy class containing C_n^2 elements, and this is the required subset.

If $n = 6$, multiplying all elements in the conjugacy class of matrices with two (-1) entries on the diagonal gives $-I$, since this class contains $C_5^1 = 5$ matrices with (-1) in a fixed position. Multiplying all elements in the conjugacy class of matrices with four (-1) entries gives the identity matrix, since exactly $C_5^3 = 10$ matrices in this conjugacy class have (-1) in a fixed position. Thus, for $n = 6$, we can define those matrices in the system $\mathcal{D}$ in which (-1) occurs exactly 2 times on the diagonal.

If $n \geq 8$, then among the matrices with two (-1) entries on the diagonal, there are always three such that the product of the first two equals the third. However, multiplying any two matrices with (-1) occurring $(n-2)$ times on the diagonal gives a matrix from a different conjugacy class. $\square$

Lemma 28. Let K and L be infinite fields of characteristic different from 2, let n be even, $n \geq 4$, and suppose that $\text{SL}_n(K) \equiv_{\forall} \text{SL}_n(L)$. Then under any finite partial isomorphism $\Phi: \text{SL}_n(K) \rightarrow \text{SL}_n(L)$ such that $\text{dom } \Phi$ contains all matrices in the system $\mathcal{D}$, a diagonal matrix is mapped to a diagonal matrix.

Proof. Since a diagonal matrix commutes with all matrices in the system $\mathcal{D}$, its image commutes with the images of the matrices in $\mathcal{D}$. We have already shown that the matrices in the system $\mathcal{D}$ can be defined by an existential formula, so their images retain their form. Therefore, the image of our matrix commutes with all matrices in the system $\mathcal{D}$ and is consequently diagonal. $\square$

Lemma 29. Let K and L be infinite fields of characteristic different from 2, let n be even, $n \geq 4$, and suppose that $\text{SL}_n(K) \equiv_{\forall} \text{SL}_n(L)$. Then under any finite partial isomorphism $\Phi: \text{SL}_n(K) \rightarrow \text{SL}_n(L)$ such that $\text{dom } \Phi$ contains all matrices in the system $\mathcal{D}$ and a finite set of auxiliary matrices from the previous lemmas, the matrix $I - E_{11} - E_{22} - E_{12} + E_{21}$ is mapped to a matrix

that, with respect to some basis (relative to which the form of the matrices in the system $\mathcal{D}$ remains unchanged), has the form

$$\begin{pmatrix} 0 & -1 & 0 & \dots & 0 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & \pm 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \pm 1 \end{pmatrix}.$$

Proof. Introduce notation for the matrices in the system $\mathcal{D}$ that have exactly two (-1) entries on the diagonal: let A_{ij} , $i < j$, denote the matrix with (-1) in positions i and j .

Fix a matrix with two (-1) entries on the diagonal; with respect to some basis, this is A_{12} . The matrices A_{ij} , $2 < i < j$, are characterised as follows: they have two (-1) entries on the diagonal, and their product with A_{12} is a matrix in $\mathcal{D}$ that is not conjugate to A_{12} . Since the matrix $I - E_{11} - E_{22} - E_{12} + E_{21}$ commutes with all matrices A_{ij} , $2 < i < j$, and with A_{12} , the same is true of its image. Denote this image by X . Consequently,

$$X = \begin{pmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & \lambda_3 & e \\ 0 & 0 & f & \lambda_4 \end{pmatrix} \text{ for } n = 4 \text{ and } X = \begin{pmatrix} a & b & 0 & \dots & 0 \\ c & d & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \lambda_n \end{pmatrix} \text{ for } n > 4.$$

Note that $(I - E_{11} - E_{22} - E_{12} + E_{21})A_{13}(I - E_{11} - E_{22} - E_{12} + E_{21})^{-1} = A_{23}$. In the system $\mathcal{D}$, fix a matrix with two (-1) entries whose product with A_{12} has two (-1) entries, and call it A_{13} ; then $A_{23} = A_{12}A_{13}$. For $n = 4$, this relation gives the following for the second block:

$$\begin{pmatrix} \lambda_3 & e \\ f & \lambda_4 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \lambda_3 & e \\ f & \lambda_4 \end{pmatrix},$$

which gives $e = f = 0$.

Further (for all $n \geq 4$), $(I - E_{11} - E_{22} - E_{12} + E_{21})^2 = A_{12}$, so $X^2 = A_{12}$ and $\lambda_i = \pm 1$, $i = 3, \dots, n$.

Returning to the previous relation, we now write it for the upper-left block:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \left(\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

What we have proved above implies that $ad - bc = \pm 1$. First consider the case $ad - bc = 1$. Then

$$\begin{pmatrix} -ad - bc & 2ab \\ -2cd & ad + bc \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

so $ad = ab = cd = 0$, $bc = -1$, and finally $a = d = 0$. Changing the basis using the matrix $I + (-1 - \frac{1}{b})E_{11}$, which commutes with all diagonal matrices, gives the required form of X .

It remains to show that the case $ad - bc = -1$ is impossible. As in the previous case, we obtain $a = d = 0$, $bc = 1$, but then

$$\begin{pmatrix} 0 & b \\ \frac{1}{b} & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

that is, $X^2 = I$, which contradicts the condition $X^2 = A_{12}$. $\square$

Lemma 30. Let K and L be infinite fields of characteristic different from 2, let n be even, $n \geq 4$, and suppose that $\text{SL}_n(K) \equiv_{\forall} \text{SL}_n(L)$. Then under any finite partial isomorphism $\Phi: \text{SL}_n(K) \rightarrow \text{SL}_n(L)$ such that $\text{dom } \Phi$ contains all matrices in the system $\mathcal{D}$, the matrix $I - E_{11} - E_{22} - E_{12} + E_{21}$, the matrix $\text{diag}[2, 1, \frac{1}{2}, 1, \dots, 1]$, and a finite set of auxiliary matrices from the previous lemmas, a matrix of the form $I + \alpha E_{12}$ is mapped to a matrix of the same form.

Proof. The proof is analogous to that of Lemma 20, with the role of $-I + E_{11} + E_{12} + E_{21} + E_{22}$ now played by the matrix $I - E_{11} - E_{22} - E_{12} + E_{21}$. $\square$

Lemma 31. Let K and L be infinite fields of characteristic different from 2, let n be even, $n \geq 4$, and suppose that $\text{SL}_n(K) \equiv_{\forall} \text{SL}_n(L)$. Then for any finite partial isomorphism $\Phi: \text{SL}_n(K) \rightarrow \text{SL}_n(L)$ such that $\text{dom } \Phi$ contains all matrices in the system $\mathcal{D}$, the matrix $I - E_{11} - E_{22} - E_{12} + E_{21}$, the matrix $\text{diag}[2, 1, \frac{1}{2}, 1, \dots, 1]$, and a finite set of auxiliary matrices from the previous lemmas, we have $\Phi(I + E_{12}) = I + E_{12}$.

Proof. Since the following relation holds (again, only the upper-left blocks of the matrices are shown):

$$A_{13} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} A_{12} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} A_{13} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the previous lemma implies the following relation for the images:

$$A_{13} \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} A_{12} \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} A_{13} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix},$$

Thus,

$$\begin{pmatrix} 1 - \alpha^2 & -\alpha \\ \alpha & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & \alpha \end{pmatrix},$$

which gives $\alpha = 1$, as required. $\square$

2.3.8 Remarks completing the proof

To complete the proof of the main theorem, Theorem 1, we use the arguments given above. More precisely, we already know how to prove that addition matrices of even degree in special linear groups retain their form, and we know that multiplication matrices remain diagonal. The proof that multiplication matrices are mapped to multiplication matrices, while the correspondence between a pair of addition and multiplication matrices and a single field element is preserved, is entirely analogous to the proof of Lemma 22, taking into account the observation in the proof of Lemma 26. All that remains is to repeat, almost word for word, the argument from the proof of Theorem 6 under the assumptions of Theorem 7. Since Lemma 16 allows us to distinguish between underlying fields of characteristic 2 and those of characteristic different from 2, the proof of Theorem 1 is complete.

Chapter 3

General and special linear groups over local commutative rings with $1/2$

In this chapter we prove the following theorem:

Theorem 2. Let R_1 and R_2 be local commutative rings with $1/2$. The groups $G_n(R_1)$ and $G_m(R_2)$ ($G = GL, SL, n, m \geq 3$) are universally equivalent if and only if $n = m$ and the rings R_1 and R_2 are universally equivalent.

3.1 Preliminary remarks based on previously established results

We first prove the easier implication. If the rings R_1 and R_2 are universally equivalent and $n = m$, then clearly $G_n(R_1) \equiv_{\forall} G_n(R_2)$, since all relations between finitely many matrices in a matrix group can be rewritten in terms of finitely many elements of the scalar rings, with a correspondence between these elements. Specifically, we can rewrite each existential sentence in the language of groups as an existential sentence in the language of scalars as follows: replace each matrix with its n^2 entries, rewrite the relations between matrices as relations between elements of the scalar rings, and, for each matrix, also require the existence of n^2 elements representing its inverse and add the relations describing the products of the matrix and its inverse (for special linear groups, we must also include the conditions that the determinants are equal to one). The resulting formula is again an existential sentence. Note that the assertion remains valid when $n = m < 3$.

Next we prove the more difficult implication: universal equivalence of the

linear groups implies equality of their degrees and universal equivalence of the corresponding rings.

Note that if the linear groups are universally equivalent, then the rings R_1 and R_2 are either both finite or both infinite. For finite structures, universal equivalence coincides with isomorphism, so the theorem follows directly from [12], where it is proved that, for positive integers $n, m \geq 3$, the following conditions are equivalent:

- 1) $n = m$ and $R_1 \cong R_2$,
- 2) $\mathrm{GL}_n(R_1) \cong \mathrm{GL}_m(R_2)$,
- 3) $\mathrm{SL}_n(R_1) \cong \mathrm{SL}_m(R_2)$,

where R_1 and R_2 are local commutative rings with $1/2$.

Thus, throughout the rest of this chapter, we assume that the rings R_1 and R_2 are infinite.

The assertion that the degrees are equal follows from the fact that, in the group $G_n(R)$ ($G = \mathrm{GL}, \mathrm{SL}$) over a local commutative ring R in which two is invertible, every family of pairwise commuting involutions is simultaneously diagonalisable. For local rings, this fact was proved in [21].

From now on, we assume that $n > 2$.

3.2 Involutions

Let $\mathcal{MJ}$ denote a fixed maximal system of pairwise commuting involutions. With respect to a suitable common basis, all elements of $\mathcal{MJ}$ are diagonal matrices with diagonal entries ± 1 . Fix one such basis. Note that $\mathcal{MJ}$ splits into conjugacy classes, each consisting precisely of the matrices with the same number of diagonal entries equal to -1 with respect to the chosen basis. We now introduce notation for certain subsets of $\mathcal{MJ}$.

First consider $\mathrm{SL}_n(R)$ for odd n , and $\mathrm{GL}_n(R)$. In $\mathcal{MJ}$, the smallest conjugacy classes other than singleton classes (if any) contain n elements: these are the classes consisting of matrices in which -1 occurs 1 time or $n - 1$ times on the diagonal (only the second possibility occurs in $\mathrm{SL}_n(R)$). These conjugacy classes are defined by an existential formula. Note that a formula based on counting the elements in conjugacy classes defines each of these two classes, but does not specify which one. Denote either of these classes by $\mathcal{J}_1$; the choice does not matter.

For $\mathrm{SL}_n(R)$ with even $n \geq 4$, let $\mathcal{J}_2$ denote the following subset of $\mathcal{MJ}$: the matrices with exactly two diagonal entries equal to -1 . The subset $\mathcal{J}_2$ is defined by an existential formula in the same way as in Lemma 27.

The fact that the set of matrices $\mathcal{MJ}$ is defined by an existential formula also implies that the form of diagonal matrices is preserved under an

isomorphism of a partial structure containing all matrices in $\mathcal{MJ}$.

3.3 Matrices with a diagonal block $-E_{12} + E_{21}$

Lemma 32. Let R_1 and R_2 be infinite local commutative rings with $1/2$, and suppose that $G_n(R_1) \equiv_{\forall} G_n(R_2)$ ($G = GL, SL$). Also, let M_1 be an arbitrary finite partial structure of $G_n(R_1)$ containing the set $\mathcal{MJ}$, and let M_2 be a finite partial structure of $G_n(R_2)$ isomorphic to M_1 (its existence follows from the criterion for universal equivalence).

Then under any isomorphism

$$\Phi: M_1 \rightarrow M_2$$

the matrix $\sigma_{12} = I - E_{11} - E_{22} - E_{12} + E_{21}$ is mapped to a matrix of the form

$$\begin{pmatrix} 0 & -1 & 0 & \dots & 0 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & \pm 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \pm 1 \end{pmatrix}$$

with respect to a basis relative to which the form of the matrices in $\mathcal{MJ}$ remains unchanged.

Proof. We consider two cases.

Case 1. If $G = GL$, or $G = SL$ with odd n , we use the fact that σ_{12} commutes with all but two matrices in $\mathcal{J}_1$. Denote these two matrices by I_1 and I_2 , assuming that the diagonal entry of I_j in position j differs from all its other diagonal entries. In this case, we will use the notation $I_{ij} = I_i I_j$, $i \neq j$.

Case 2. If $G = SL$ and n is even, we proceed as follows. First fix a matrix in $\mathcal{J}_2$, denote it by I_{12} , and assume that its first and second diagonal entries are (-1) . Then consider all matrices in $\mathcal{J}_2$ whose product with I_{12} does not belong to $\mathcal{J}_2$; all these matrices, as well as I_{12} , commute with σ_{12} .

In each case, the image has the following form:

$$\Phi(\sigma_{12}) = \begin{pmatrix} d_{11} & d_{12} & 0 & 0 \\ d_{21} & d_{22} & 0 & 0 \\ 0 & 0 & d_{33} & a \\ 0 & 0 & b & d_{nn} \end{pmatrix} \text{ for } n = 4 \text{ or}$$

$$\Phi(\sigma_{12}) = \begin{pmatrix} d_{11} & d_{12} & 0 & \dots & 0 \\ d_{21} & d_{22} & 0 & \dots & 0 \\ 0 & 0 & d_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_{nn} \end{pmatrix} \text{ for even } n \neq 4.$$

Next, for $n \neq 4$, we use the relation $\sigma_{12}^2 = I_{12}$. For the images, this gives:

$$\begin{pmatrix} d_{11}^2 + d_{12}d_{21} & d_{11}d_{12} + d_{12}d_{22} & 0 & \dots & 0 \\ d_{21}d_{11} + d_{22}d_{21} & d_{21}d_{12} + d_{22}^2 & 0 & \dots & 0 \\ 0 & 0 & d_{33}^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_{nn}^2 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 & \dots & 0 \\ 0 & -1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix},$$

which, since the local ring contains $1/2$, immediately implies that $d_{ii} = \pm 1$ for all $2 < i \leq n$.

We now consider the first case separately. We use another relation: $\sigma_{12}I_1\sigma_{12} = I_1$. For the images, it gives the following (only the upper-left block is shown):

$$\begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} =$$

$$= \begin{pmatrix} -d_{11}^2 + d_{12}d_{21} & -d_{11}d_{12} + d_{12}d_{22} \\ -d_{21}d_{11} + d_{22}d_{21} & -d_{21}d_{12} + d_{22}^2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Adding and subtracting the corresponding entries of the matrices in this and the preceding relation gives $2d_{12}d_{21} = -2$, that is, $d_{12}d_{21} = -1$, as well as $2d_{22}d_{21} = 0$ and $2d_{11}d_{12} = 0$. Since d_{12}, d_{21} are invertible, it follows that $d_{11} = d_{22} = 0$.

We now consider the second case separately. Here we use the relation $\sigma_{12}I_{13}\sigma_{12}^{-1} = I_{23}$. (In $\mathcal{I}_2$, we fixed a matrix whose product with I_{12} belongs to $\mathcal{I}_2$ and denoted it by I_{13} ; then $I_{23} = I_{12}I_{13}$.) Note that for $n = 4$, this relation implies that the second block is diagonal, so, as for $n \neq 4$, its diagonal entries are ± 1 .

We now write this relation for the upper-left blocks of the images:

$$\begin{aligned} & \frac{1}{d_{11}d_{22} - d_{12}d_{21}} \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} d_{22} & -d_{12} \\ -d_{21} & d_{11} \end{pmatrix} = \\ & = \frac{1}{d_{11}d_{22} - d_{12}d_{21}} \begin{pmatrix} -d_{11}d_{22} - d_{12}d_{21} & 2d_{11}d_{12} \\ -2d_{21}d_{22} & d_{11}d_{22} + d_{12}d_{21} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned}$$

Since $d_{11}d_{12} = d_{21}d_{22} = 0$, the matrix is invertible, and the ring is local, either 1) $d_{11} = d_{22} = 0$ and d_{12} and d_{21} are invertible, or 2) $d_{12} = d_{21} = 0$ and d_{11} and d_{22} are invertible. In case 2), the entry in position $(1, 1)$ is both $-\frac{d_{11}d_{22}}{d_{11}d_{22}} = -1$ and 1 , which is impossible. Thus only case 1) can occur, and the equality in position $(1, 1)$ in the relation $\sigma_{12}^2 = I_{12}$ gives $d_{12}d_{21} = -1$.

We now know that

$$\Phi(\sigma_{12}) = \begin{pmatrix} 0 & -\alpha & 0 & \dots & 0 \\ \frac{1}{\alpha} & 0 & 0 & \dots & 0 \\ 0 & 0 & \pm 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \pm 1 \end{pmatrix},$$

where α is an invertible element of the ring. It remains to apply the change of basis given by the matrix $\text{diag}[1/\alpha, 1, \dots, 1]$, which commutes with all matrices in $\mathcal{MJ}$, to obtain the required result. $\square$

Remark 4. Exactly the same method shows that the form of the matrix σ_{23} , whose block $-E_{12} + E_{21}$ lies at the intersection of the second and third rows with the second and third columns, is preserved. Arguing as in Lemma 32, we first obtain

$$\Phi(\sigma_{23}) = \begin{pmatrix} \pm 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & -\beta & 0 & \dots & 0 \\ 0 & \frac{1}{\beta} & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \pm 1 & & \\ \vdots & \vdots & \vdots & & \ddots & \\ 0 & 0 & 0 & & & \pm 1 \end{pmatrix},$$

and then apply the change of basis given by the matrix $\text{diag}[1, 1, \beta, 1, \dots, 1]$, which commutes both with all elements of $\mathcal{MJ}$ and with σ_{12} .

3.4 Preservation of transvections

Lemma 33. Let R_1 and R_2 be infinite local commutative rings with $1/2$, and suppose that $G_n(R_1) \equiv_{\forall} G_n(R_2)$ ($G = \text{GL}, \text{SL}$). Let M_1 be an arbitrary

finite partial structure of $G_n(R_1)$ containing the finite set of matrices from the preceding lemma and the matrix $\text{diag}[2, 1, 1/2, 1, \dots, 1]$, and let M_2 be a finite partial structure of $G_n(R_2)$ isomorphic to M_1 . Then under any partial isomorphism

$$\Phi: M_1 \rightarrow M_2$$

the matrix $I + E_{12}$ is mapped, up to an automorphism of the group, to a matrix of the same form with respect to a basis relative to which the form of the matrices in $\mathcal{MJ}$, σ_{12} , and σ_{23} remains unchanged.

Proof. As in the proof of Lemma 32,

$$\Phi(I + E_{12}) = \begin{pmatrix} t_{11} & t_{12} & 0 & \dots & 0 \\ t_{21} & t_{22} & 0 & \dots & 0 \\ 0 & 0 & t_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & t_{nn} \end{pmatrix}.$$

Since $\text{diag}[2, 1, 1/2, 1, \dots, 1]$ commutes with every matrix in $\mathcal{MJ}$, we have $\Phi(\text{diag}[2, 1, 1/2, 1, \dots, 1]) = \text{diag}[\alpha_1, \dots, \alpha_n]$, where all α_i are invertible. Applying the relation $\text{diag}[2, 1, 1/2, 1, \dots, 1](I + E_{12})(\text{diag}[2, 1, 1/2, 1, \dots, 1])^{-1} = (I + E_{12})^2$ to the images gives $t_{ii}^2 = t_{ii}$ for all $3 \leq i \leq n$. Since all t_{ii} with $3 \leq i \leq n$ are invertible, they are all equal to one.

In the calculations, we can now consider only the upper-left blocks of size 2, or, where necessary, of size 3. We use three relations:

$$((I + E_{12})I_{23})^2 = I,$$

$$(\sigma_{12}(I + E_{12}))^3 = I_{12},$$

$$I + E_{12} \text{ commutes with } \sigma_{23}(I + E_{12})\sigma_{23}^{-1}.$$

(In particular, the second relation implies that all diagonal entries of $\Phi(\sigma_{12})$ from the third position onwards are equal to one.)

We write these relations for the images, using the previously established preservation of the form of certain matrices. Note that the first diagonal entry of $\Phi(\sigma_{23})$ is one, since σ_{12} and σ_{23} are conjugate:

$$\begin{aligned}
\left(\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right)^2 &= \begin{pmatrix} t_{11}^2 - t_{12}t_{21} & t_{12}(-t_{11} + t_{22}) \\ t_{21}(t_{11} - t_{22}) & t_{22}^2 - t_{12}t_{21} \end{pmatrix} = I, \\
\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} \right)^3 &= \begin{pmatrix} -t_{21} & -t_{22} \\ t_{11} & t_{12} \end{pmatrix}^3 = \\
&= \begin{pmatrix} -t_{21}^3 + 2t_{11}t_{21}t_{22} - t_{11}t_{12}t_{22} & -t_{21}^2t_{22} + t_{11}t_{22}^2 + t_{12}t_{21}t_{22} - t_{12}^2t_{22} \\ * & t_{11}t_{21}t_{22} - 2t_{11}t_{12}t_{22} + t_{12}^3 \end{pmatrix} = -I,
\end{aligned}$$

$$\begin{aligned}
&\begin{pmatrix} t_{11} & t_{12} & 0 \\ t_{21} & t_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} t_{11} & t_{12} & 0 \\ t_{21} & t_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = \\
&= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} t_{11} & t_{12} & 0 \\ t_{21} & t_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} t_{11} & t_{12} & 0 \\ t_{21} & t_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\
&\text{or, equivalently, } \begin{pmatrix} t_{11}^2 & t_{12} & t_{11}t_{12} \\ t_{11}t_{21} & t_{22} & t_{12}t_{21} \\ t_{21} & 0 & t_{22} \end{pmatrix} = \begin{pmatrix} t_{11}^2 & t_{11}t_{12} & t_{12} \\ t_{21} & t_{22} & 0 \\ t_{11}t_{21} & t_{12}t_{21} & t_{22} \end{pmatrix}.
\end{aligned}$$

The last relation gives $t_{12}t_{21} = 0$, so the first yields $t_{11}^2 = t_{22}^2 = 1$. Taking determinants in the second gives $1 = (t_{11}t_{22} - t_{12}t_{21})^3 = (t_{11}t_{22})^3 = t_{11}^2t_{22}^2t_{11}t_{22} = t_{11}t_{22}$. The same relation gives $t_{21}^2t_{22} - t_{12}t_{21}t_{22} + t_{12}^2t_{22} = t_{11}t_{22}^2$; using $t_{12}t_{21} = 0$ and the invertibility of t_{22} , we obtain $t_{12}^2 + t_{21}^2 = t_{11}t_{22}$. Since the ring is local, one of t_{12} and t_{21} is invertible, and the other is therefore zero.

If $t_{21} = 0$, then $t_{11}t_{12}t_{22} = 1$ (by the second relation), so $t_{12} = 1$. Next, $t_{12} = t_{11}t_{12}$ (by the third relation) gives $t_{11} = 1$, and finally $t_{22} = 1$, as required.

If $t_{12} = 0$, then the second relation again gives $t_{11}t_{21}t_{22} = -1$, so $t_{21} = -1$, and then $t_{11} = 1 = t_{22}$. It remains to apply the contragredient automorphism, which maps each group element A to $(A^T)^{-1}$ and fixes all elements of $\mathcal{MJ}$, as well as σ_{12} and σ_{23} . $\square$

Lemma 34. Let R_1 and R_2 be infinite local commutative rings with $1/2$, and suppose that $G_n(R_1) \equiv_{\forall} G_n(R_2)$ ($G = \text{GL}, \text{SL}$). Let M_1 be an arbitrary finite partial structure of $G_n(R_1)$ containing the finite set of matrices from the preceding lemmas and the matrix $\text{diag}[2, 1, 1/2, 1, \dots, 1]$, and let M_2 be a

finite partial structure of $G_n(R_2)$ isomorphic to M_1 . Then under any partial isomorphism

$$\Phi: M_1 \rightarrow M_2$$

any finite set of matrices of the form $I + \alpha_i E_{12}$, $i = 1, \dots, k$ is mapped to a set of matrices of the form $I + \beta_i E_{12}$, $i = 1, \dots, k$, with respect to a basis relative to which the form of the matrices from the preceding lemmas remains unchanged.

Proof. As in Lemma 33, it is easy to show that

$$\Phi(I + \alpha_i E_{12}) = \begin{pmatrix} s_{11}^i & s_{12}^i & 0 \\ s_{21}^i & s_{22}^i & 0 \\ 0 & 0 & I \end{pmatrix}.$$

Next, since the matrices $I + E_{12}$ and $I + \alpha_i E_{12}$ commute, we obtain $s_{21}^i = 0$, $s_{11}^i = s_{22}^i$.

It remains to show that the first two diagonal entries are one. We write out the relations involving the diagonal matrix for the upper-left block, using $\Phi(\text{diag}[2, 1, 1/2, 1, \dots, 1]) = \text{diag}[d_1, \dots, d_n]$:

$$\begin{pmatrix} s_{11}^i & \frac{d_1}{d_2} s_{12}^i \\ 0 & s_{22}^i \end{pmatrix} = \begin{pmatrix} (s_{11}^i)^2 & 2s_{11}^i s_{12}^i \\ 0 & (s_{22}^i)^2 \end{pmatrix}.$$

The invertibility of s_{11}^i, s_{22}^i now implies that both elements are equal to one. $\square$

We make two remarks on the connection between multiplication of elements of the local ring R and operations on elements of the group $G_n(R)$ ($G = \text{GL}, \text{SL}$). First,

$$[I + \alpha E_{12}, I + \beta E_{23}] = I + \alpha\beta E_{13}.$$

Second,

$$\begin{aligned} I + \alpha E_{23} &= \sigma_{12} \sigma_{23}^{-1} (I + \alpha E_{12})^{-1} \sigma_{23} \sigma_{12}^{-1}, \\ I + \alpha E_{13} &= [I + E_{12}, I + \alpha E_{23}]. \end{aligned}$$

The main assertion now follows from the preceding lemmas, just as in the previous chapter. More explicitly, let $S_1 \subset R_1$ be a finite partial structure of the ring R_1 . Our task is to find a partial structure S_2 of R_2 that is partially isomorphic to it. Let $M_1 \subset G_n(R_1)$ ($G = \text{GL}, \text{SL}$) be a finite partial structure containing all matrices of the form $I + \alpha E_{12}$ for all $\alpha \in S_1$, together with finitely many auxiliary matrices from the preceding lemmas. Since the groups are universally equivalent, the partial structure M_1 has an isomorphic

partial structure $M_2 \subset G_n(R_2)$ ($G = GL, SL$), and every isomorphism $\Phi: M_1 \rightarrow M_2$ preserves the form of the matrices. Thus, with respect to a suitable common basis, the images of the matrices $I + \alpha E_{12}, I + \alpha E_{13}, I + \alpha E_{23}$ have the form $I + \beta_1 E_{12}, I + \beta_2 E_{13}, I + \beta_3 E_{23}$, with $\beta_1 = \beta_2 = \beta_3 \in R_2$. Then S_2 is the set of all elements of R_2 corresponding to the triples of images of the matrices $I + \alpha E_{12}, I + \alpha E_{13}, I + \alpha E_{23}$ for all elements $\alpha \in S_1$.

Chapter 4

General linear groups over noncommutative local rings with $1/2$

In this chapter we prove the following theorem:

Theorem 3. Let R_1 and R_2 be local (not necessarily commutative) rings with $1/2$. The groups $\mathrm{GL}_n(R_1)$ and $\mathrm{GL}_m(R_2)$ ($n, m \geq 3$) are universally equivalent if and only if either $n = m$ and the rings R_1 and R_2 are universally equivalent, or $n = m$ and the rings R_1 and R_2^{op} are universally equivalent.

4.1 The easier implication: equivalence of rings implies equivalence of groups

The proof of the easier implication—that universal equivalence of the rings implies universal equivalence of the general linear groups—is entirely analogous to the proof for commutative local rings given in Chapter 3, and likewise remains valid when $n = m < 3$.

If the rings R_1 and R_2^{op} are universally equivalent, then the groups $\mathrm{GL}_n(R_1)$ and $\mathrm{GL}_n(R_2^{op})$ are universally equivalent. Now note that $\mathrm{GL}_n(R_2^{op}) \cong \mathrm{GL}_n(R_2)^{op}$ (with transposition as the isomorphism), and $\mathrm{GL}_n(R_2)^{op} \cong \mathrm{GL}_n(R_2)$ (with inversion as the isomorphism). Thus $\mathrm{GL}_n(R_1)$ and $\mathrm{GL}_n(R_2)$ are universally equivalent in this case as well.

4.2 The more difficult implication: from groups to rings

We now turn to the proof of the more difficult implication. Note that if the linear groups are universally equivalent, then the rings R_1 and R_2 are either both finite or both infinite. For finite structures, universal equivalence coincides with isomorphism and elementary equivalence, so the theorem follows from [5]. Specifically, Theorem 4 of that paper states that, in the absence of nontrivial central idempotents (which holds under the assumptions of Theorem 3, since the rings are local), the matrix rings satisfy $M_n(R_1) \cong M_m(R_2)$ or $M_n(R_1) \cong M_m(R_2)^{op}$. In the first case, since the rings are local, an isomorphism of the matrix rings implies an isomorphism of the scalar rings (Corollary 17.31 in [16]). In the second case, we first use the isomorphism $M_m(R_2)^{op} \cong M_m(R_2^{op})$ (transposition), and then apply the same corollary. We will prove equality of the dimensions below, treating finite and infinite rings together.

4.2.1 Determining the dimension

We first study the structure of involutions in the general linear group over a noncommutative local ring, following [21], where analogous arguments are given for local commutative rings.

Lemma 35. Let R be a local ring with $1/2$, $J \in GL_n(R)$, and $J^2 = I$. Then the right R -module V of rank n has a basis with respect to which J is represented by a matrix of the form $\text{diag}[1, \dots, 1, -1, \dots, -1]$, with t ones and s minus ones on the diagonal, where $t + s = n$ and t, s are uniquely determined.

Proof. Consider the following submodules:

$$N(J) = \{v \in V \mid J(v) = -v\}, \quad P(J) = \{v \in V \mid J(v) = v\}.$$

Let $u \in N(J) \cap P(J)$. Then $J(u) = u = -u$, and since the ring contains $1/2$, it follows that $u = 0$.

Now let u be an arbitrary element of V . We have

$$u = (u - J(u))\frac{1}{2} + (u + J(u))\frac{1}{2},$$

where $u - J(u) \in N(J)$, $u + J(u) \in P(J)$. Hence $V = N(J) \oplus P(J)$.

It remains to make two remarks on the structure of the submodules $N(J)$ and $P(J)$. First, $N(J)$ and $P(J)$ are projective modules, since they are direct

summands of a free module. Second, since projective modules over a local (not necessarily commutative) ring are free ([15], Theorem 19.29), $N(J)$ and $P(J)$ are free modules whose ranks are finite and uniquely determined. With respect to a basis obtained by combining bases of the submodules $P(J)$ and $N(J)$, the matrix of J has the required form. $\square$

Lemma 36. Let R be a local ring with $1/2$, $I_1, \dots, I_m \in \text{GL}_n(R)$, $I_j^2 = I$, and $I_j I_k = I_k I_j$ for all $j, k = 1, \dots, m$. Then, with respect to a suitable basis of the right R -module of rank n , all $I_1, \dots, I_m$ are diagonal with diagonal entries ± 1 .

Proof. First consider two commuting operators I_1, I_2 of order two. By Lemma 35, we may assume that a basis of the module has already been fixed with respect to which $I_1 = \text{diag}[1, \dots, 1, -1, \dots, -1]$ (t ones and s minus ones). Since $I_1 I_2 = I_2 I_1$ and two is invertible, I_2 has block diagonal form with respect to the chosen basis, with diagonal blocks of sizes $t \times t$ and $s \times s$.

Let $v \in P(I_1)$. Then $I_1 I_2(v) = I_2 I_1(v) = I_2(v)$, so $I_2(v) \in P(I_1)$; that is, the submodule $P(I_1)$ is invariant under I_2 . The same holds for the submodule $N(I_1)$. It now suffices to change the basis in each of the submodules $P(I_1)$ and $N(I_1)$ so that I_2 takes the required form, while the form of I_1 remains unchanged.

For more operators, the argument continues by induction. Suppose that for $I_1, \dots, I_k$ we have already constructed a basis with respect to which all operators are diagonal with diagonal entries ± 1 . The entire module is then a direct sum of all possible intersections of the invariant submodules $P(I_j)$, $j = 1, \dots, k$ and $N(I_j)$, $j = 1, \dots, k$. Since I_{k+1} commutes with each of $I_1, \dots, I_k$, all direct summands are invariant under I_{k+1} . Thus I_{k+1} can be diagonalised on each direct summand, as required. $\square$

Corollary 1. Let R be a local ring with $1/2$. The cardinality of a maximal subset of $\text{GL}_n(R)$ consisting of pairwise commuting involutions is $2^n - 1$.

Corollary 2. Let R_1 and R_2 be local rings with $1/2$, let m and n be positive integers, and suppose that the groups $\text{GL}_n(R_1)$ and $\text{GL}_m(R_2)$ are universally equivalent. Then $n = m$.

Throughout the rest of this chapter, we assume that $n > 2$ and that the rings R_1 and R_2 are infinite.

4.2.2 Elements of order 2

Fix a maximal subset of pairwise commuting involutions and a basis of the module with respect to which all these involutions are diagonal. Denote this

set of involutions by $\mathcal{MJ}$. Note that $\mathcal{MJ}$ splits into conjugacy classes, each consisting precisely of the elements with the same number of diagonal entries equal to -1 . The conjugacy class of smallest cardinality is $\{-I\}$, followed by the n -element conjugacy classes consisting of matrices with exactly one minus one or $n - 1$ minus ones on the diagonal. These conjugacy classes are defined by an existential formula. Note that a formula based on counting the elements in a conjugacy class defines each of these two classes, but does not specify which one. Denote the class of matrices with one minus one by $\mathcal{J}_1$, and the class with a single diagonal entry equal to one by $\mathcal{J}_2$. For the elements of $\mathcal{J}_1$, introduce the following notation: I_k is the matrix whose diagonal entry in position k is minus one.

Note that under any isomorphism of a partial structure containing $\mathcal{MJ}$, this set is mapped to a set of elements that, with respect to a suitable common basis, have the same form as $\mathcal{MJ}$ with respect to the previously fixed basis; the set $\mathcal{J}_1$ is mapped either to a set of the same form or to $\mathcal{J}_2$. If the domain also contains a diagonal matrix, its image is diagonal as well, since the diagonal matrices are precisely those that commute with every element of $\mathcal{MJ}$.

We will use the following auxiliary lemma.

Lemma 37. Let R be a local ring with $1/2$, $x \in R$, and $x^2 = 1$. Then $x = \pm 1$.

Proof. The element $\frac{1}{2}(x + 1)$ is idempotent. Since the ring R is local, this idempotent is trivial: $\frac{1}{2}(x + 1) = 0$ or 1 . In the first case $x = -1$, and in the second $x = 1$. $\square$

We now prove that the form of the matrices we need is preserved under isomorphisms of partial structures.

Lemma 38. Let R_1 and R_2 be infinite local rings with $1/2$, and suppose that $\text{GL}_n(R_1) \equiv_{\forall} \text{GL}_n(R_2)$ ($n > 2$). Also, let M_1 be an arbitrary finite partial structure of $\text{GL}_n(R_1)$ containing the set $\mathcal{MJ}$ and matrices conjugating its elements, and let M_2 be a finite partial structure of $\text{GL}_n(R_2)$ isomorphic to M_1 (its existence follows from the criterion for universal equivalence).

Then under any isomorphism

$$\Phi: M_1 \rightarrow M_2$$

the matrix $\sigma_{12} = I - E_{11} - E_{22} + E_{12} + E_{21}$ is mapped to a matrix of the

form

$$\begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & \pm 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & & & \pm 1 \end{pmatrix}$$

with respect to a basis relative to which the form of the matrices in $\mathcal{MJ}$ remains unchanged.

Proof. Write the image as $\Phi(\sigma_{12}) = (d_{ij})$.

Since σ_{12} satisfies the following three relations:

$$\text{for } k = 3, \dots, n \ (\sigma_{12}I_k = I_k\sigma_{12}), \quad (4.1)$$

$$\sigma_{12}^2 = I, \quad (4.2)$$

$$\sigma_{12}I_1\sigma_{12} = I_2, \quad (4.3)$$

the images satisfy the corresponding relations.

The first relation implies that

$$\Phi(\sigma_{12}) = \begin{pmatrix} d_{11} & d_{12} & 0 & \dots & 0 \\ d_{21} & d_{22} & 0 & \dots & 0 \\ 0 & 0 & d_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_{nn} \end{pmatrix},$$

and the second relation gives $d_{kk} = \pm 1$ for $3 \leq k \leq n$.

We write out the second and third relations for the upper-left blocks (up to multiplying both sides of the equality by $-I$):

$$\begin{aligned} \begin{pmatrix} d_{11}^2 + d_{12}d_{21} & d_{11}d_{12} + d_{12}d_{22} \\ d_{21}d_{11} + d_{22}d_{21} & d_{21}d_{12} + d_{22}^2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \\ \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} &= \\ = \begin{pmatrix} -d_{11}^2 + d_{12}d_{21} & -d_{11}d_{12} + d_{12}d_{22} \\ -d_{21}d_{11} + d_{22}d_{21} & -d_{21}d_{12} + d_{22}^2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned}$$

Adding the entries in position $(1, 1)$ and subtracting the entries in position $(2, 2)$ in the equalities above gives $2d_{12}d_{21} = 2$ and $2d_{21}d_{12} = 2$. Thus $d_{12} = \alpha$, $d_{21} = 1/\alpha$, where α is an invertible element of the ring R_2 .

Adding the entries in position $(1, 2)$ gives $2d_{12}d_{22} = 0$. The element $2d_{12}$ is invertible, so $d_{22} = 0$; similarly, $d_{11} = 0$.

We now apply the change of basis given by the invertible matrix

$$\text{diag}[1/\alpha, 1, \dots, 1],$$

which commutes with all matrices in $\mathcal{MJ}$, thereby obtaining the required form for $\Phi(\sigma_{12})$. $\square$

Corollary 3. Let R_1 and R_2 be infinite local rings with $1/2$, and suppose that $\text{GL}_n(R_1) \equiv_{\forall} \text{GL}_n(R_2)$ ($n > 2$). Also, let M_1 be an arbitrary finite partial structure of $\text{GL}_n(R_1)$ containing the set $\mathcal{MJ}$ and matrices conjugating its elements, and let M_2 be a finite partial structure of $\text{GL}_n(R_2)$ isomorphic to M_1 .

Then under any isomorphism

$$\Phi: M_1 \rightarrow M_2$$

the matrix $\sigma_{23} = I - E_{22} - E_{33} + E_{23} + E_{32}$ is mapped to a matrix of the form

$$\begin{pmatrix} \pm 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \pm 1 & & \\ \vdots & \vdots & \vdots & & \ddots & \\ 0 & 0 & 0 & & & \pm 1 \end{pmatrix}$$

with respect to a basis relative to which the form of the matrices in $\mathcal{MJ}$ remains unchanged.

Proof. Arguing as in the proof of the preceding lemma, we can show that

$$\Phi(\sigma_{23}) = \begin{pmatrix} \pm 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & \beta & 0 & \dots & 0 \\ 0 & \frac{1}{\beta} & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \pm 1 & & \\ \vdots & \vdots & \vdots & & \ddots & \\ 0 & 0 & 0 & & & \pm 1 \end{pmatrix},$$

where β is an invertible element of the ring R_2 . It remains to apply the change of basis given by the matrix $\text{diag}[1, 1, \beta, 1, \dots, 1]$, which commutes with every matrix in $\mathcal{MJ}$ and with σ_{12} . $\square$

4.2.3 The matrix $I + E_{12}$: the common part of the argument

Lemma 39. Let R_1 and R_2 be infinite local rings with $1/2$, and suppose that $\text{GL}_n(R_1) \equiv_{\vee} \text{GL}_n(R_2)$ ($n > 2$). Let M_1 be an arbitrary finite partial structure of $\text{GL}_n(R_1)$ containing the finite set of matrices from the preceding lemma and corollary, as well as $\text{diag}[2, 1, 1/2, 1, \dots, 1]$, and let M_2 be a finite partial structure of $\text{GL}_n(R_2)$ isomorphic to M_1 . Then under any isomorphism

$$\Phi: M_1 \rightarrow M_2$$

the matrix $I + E_{12}$ is mapped to a matrix of the form $I + E_{12}$ or $I - E_{21}$, with respect to a basis relative to which the form of the matrices in $\mathcal{MJ}$, σ_{12} , and σ_{23} remains unchanged, or with respect to a basis obtained from that basis by conjugation by $\text{diag}[-1, 1, -1, 1, \dots, 1]$.

Proof. As in the proof of Lemma 38,

$$\Phi(I + E_{12}) = \begin{pmatrix} t_{11} & t_{12} & 0 & \dots & 0 \\ t_{21} & t_{22} & 0 & \dots & 0 \\ 0 & 0 & t_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & t_{nn} \end{pmatrix}.$$

Since $D = \text{diag}[2, 1, 1/2, 1, \dots, 1]$ commutes with every matrix in $\mathcal{MJ}$, we have $\Phi(\text{diag}[2, 1, 1/2, 1, \dots, 1]) = \text{diag}[\alpha_1, \dots, \alpha_n]$, where all α_i are invertible.

We list all relations needed for the rest of the proof that hold in M_1 (the corresponding relations hold for the images under the isomorphism in M_2):

$$D(I + E_{12})D^{-1} = (I + E_{12})^2, \quad (4.4)$$

$$(I_2(I + E_{12}))^2 = I, \quad (4.5)$$

$$((I_2\sigma_{12})(I + E_{12}))^3 = I, \quad (4.6)$$

$$(I_3\sigma_{23})(I + E_{12})(I_3\sigma_{23})^{-1} \text{ commutes with } I + E_{12}. \quad (4.7)$$

Note that relation (4.5) implies $t_{kk}^2 = 1$ for the diagonal entries (for $k = 3, \dots, n$). Hence $t_{kk} = \pm 1$ by Lemma 37, and these entries are, in particular, central elements of the ring.

Relation (4.4) then gives $\alpha_k t_{kk} \alpha_k^{-1} = t_{kk} = t_{kk}^2 = 1$ for $k = 3, \dots, n$.

First consider the case where the image of I_2 has exactly one minus one on the diagonal. In this case, the image of I_3 also has exactly one minus one. Relation (4.6) then implies that all diagonal entries of the image of σ_{12} from

the third position onwards are one. Next, the relation $(\sigma_{12}\sigma_{23})^3 = I$ implies that all diagonal entries of the image of σ_{23} , except the second and third, are one.

We now write relations (4.4) – (4.7) for the upper-left blocks:

$$\begin{pmatrix} t_{11}^2 + t_{12}t_{21} & t_{11}t_{12} + t_{12}t_{22} \\ t_{21}t_{11} + t_{22}t_{21} & t_{21}t_{12} + t_{22}^2 \end{pmatrix} = \begin{pmatrix} \alpha_1 t_{11} \alpha_1^{-1} & \alpha_1 t_{12} \alpha_2^{-1} \\ \alpha_2 t_{21} \alpha_1^{-1} & \alpha_2 t_{22} \alpha_2^{-1} \end{pmatrix}, \quad (4.8)$$

$$\begin{pmatrix} t_{11}^2 - t_{12}t_{21} & t_{11}t_{12} - t_{12}t_{22} \\ -t_{21}t_{11} + t_{22}t_{21} & -t_{21}t_{12} + t_{22}^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (4.9)$$

$$\begin{pmatrix} t_{21}^3 - t_{22}t_{11}t_{21} - t_{21}t_{22}t_{11} + t_{22}t_{12}t_{11} & t_{21}^2t_{22} - t_{22}t_{11}t_{22} - t_{21}t_{22}t_{12} + t_{22}t_{12}^2 \\ -t_{11}t_{21}^2 + t_{12}t_{11}t_{21} + t_{11}t_{22}t_{11} - t_{12}^2t_{11} & -t_{11}t_{21}t_{22} + t_{12}t_{11}t_{22} + t_{11}t_{22}t_{12} - t_{12}^3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (4.10)$$

$$\begin{pmatrix} t_{11}^2 & t_{11}t_{12} & t_{12} \\ t_{21} & t_{22} & 0 \\ t_{21}t_{11} & t_{21}t_{12} & t_{22} \end{pmatrix} = \begin{pmatrix} t_{11}^2 & t_{12} & t_{11}t_{12} \\ t_{21}t_{11} & t_{22} & t_{21}t_{12} \\ t_{21} & 0 & t_{22} \end{pmatrix}. \quad (4.11)$$

From (4.11) we obtain $t_{21}t_{12} = 0$. Substituting this into the entry in position (2, 2) in (4.9) gives $t_{22}^2 = 1$, so $t_{22} = \pm 1$ belongs to the centre of the ring R_2 . The entry in position (2, 2) in relation (4.8) now gives $t_{22}^2 = \alpha_2 t_{22} \alpha_2^{-1} = t_{22}$, hence $t_{22} = 1$.

We consider two cases: 1) t_{12} is invertible, and 2) t_{11} is invertible (at least one of these conditions holds, since $\Phi(I + E_{12})$ is invertible and the ring is local).

Case 1: t_{12} is invertible.

Returning to (4.9), the entry in position (1, 2) gives $(t_{11} - 1)t_{12} = 0$, so $t_{11} = 1$. The entry in position (1, 1) in (4.8) then gives $1 + t_{12}t_{21} = 1$, hence $t_{21} = 0$.

To prove that $t_{12} = 1$ in this case, we use the entry in position (1, 1) in relation (4.10): $1 = t_{21}^3 - t_{22}t_{11}t_{21} - t_{21}t_{22}t_{11} + t_{22}t_{12}t_{11} = t_{12}$.

Case 2: t_{11} is invertible. Substitute $t_{22} = 1$ and $t_{21}t_{12} = 0$ into the entry in position (1, 2) in relation (4.10):

$$t_{21}^2t_{22} - t_{22}t_{11}t_{22} - t_{21}t_{22}t_{12} + t_{22}t_{12}^2 = t_{21}^2 - t_{11} + t_{12}^2 = 0,$$

Since t_{11} is invertible, the equality $t_{21}^2 + t_{12}^2 = t_{11}$ implies that at least one of t_{12}, t_{21} is invertible. If t_{12} is invertible, the result has already been proved above. Now suppose that t_{21} is invertible. Equality (4.11) then implies $t_{11} = 1$ and $t_{12} = 0$ (positions (3, 1) and (3, 2), respectively). We determine t_{21} . The entries in positions (1,1) and (1,2) in (4.10) give

$$t_{21}^3 - 2t_{21} = 1, \quad -t_{21}^2 + 1 = 0,$$

which yields $t_{21} = -1$.

Now consider the case where the image of I_2 has exactly one diagonal entry equal to one. In this case, the image of I_3 also has exactly one diagonal entry equal to one. Relation (4.6) then implies that all diagonal entries of the image of σ_{12} from the third position onwards are minus one. Next, the relation $(\sigma_{12}\sigma_{23})^3 = I$ implies that all diagonal entries of the image of σ_{23} , except the second and third, are minus one. In this case, we first apply the change of basis given by $\text{diag}[-1, 1, -1, 1, \dots, 1]$. This matrix commutes with all diagonal involutions, so their form remains unchanged. The upper-left blocks of the images of σ_{12} and σ_{23} take the forms

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

respectively. Conditions (4.8) – (4.11) remain unchanged, so the image of $I + E_{12}$ has the stated form in this case as well. $\square$

We now divide the proof into two cases:

1) $\Phi(I + E_{12}) = I + E_{12}$

or

2) $\Phi(I + E_{12}) = I - E_{21}$

with respect to a basis relative to which the form of the matrices in $\mathcal{MJ}$, σ_{12} , and σ_{23} remains unchanged.

4.2.4 Case 1: $\Phi(I + E_{12}) = I + E_{12}$

Lemma 40. Let R_1 and R_2 be infinite local rings with $1/2$, and suppose that $\text{GL}_n(R_1) \equiv_{\forall} \text{GL}_n(R_2)$. Let M_1 be an arbitrary finite partial structure of $\text{GL}_n(R_1)$ containing the finite set of matrices from the preceding lemmas and the matrix $\text{diag}[2, 1, 1/2, 1, \dots, 1]$, and let M_2 be a finite partial structure of $\text{GL}_n(R_2)$ isomorphic to M_1 . Then under any partial isomorphism

$$\Phi: M_1 \rightarrow M_2,$$

such that $\Phi(I + E_{12}) = I + E_{12}$, any finite set of matrices of the form $I + \alpha_i E_{12}$, $i = 1, \dots, k$ is mapped to a set of matrices of the form $I + \beta_i E_{12}$, $i = 1, \dots, k$, with respect to a basis relative to which the form of the matrices from the preceding lemmas remains unchanged.

Proof. As in Lemma 39 (using commutativity with the matrices $I_3, \dots, I_n$ and relations analogous to (4.4) and (4.5)), it is easy to show that

$$\Phi(I + \alpha_i E_{12}) = \begin{pmatrix} s_{11}^i & s_{12}^i & 0 \\ s_{21}^i & s_{22}^i & 0 \\ 0 & 0 & I \end{pmatrix}.$$

Next, since the matrices $I + E_{12}$ and $I + \alpha_i E_{12}$ commute, we obtain $s_{11}^i = s_{22}^i$ and $s_{21}^i = 0$.

It remains to show that the first two diagonal entries are one. We write out the relations involving the diagonal matrix for the upper-left block, using $\Phi(\text{diag}[2, 1, 1/2, 1, \dots, 1]) = \text{diag}[d_1, \dots, d_n]$:

$$\begin{pmatrix} d_1 s_{11}^i d_1^{-1} & d_1 s_{12}^i d_2^{-1} \\ 0 & d_2 s_{11}^i d_2^{-1} \end{pmatrix} = \begin{pmatrix} (s_{11}^i)^2 & s_{11}^i s_{12}^i + s_{12}^i s_{11}^i \\ 0 & (s_{11}^i)^2 \end{pmatrix}.$$

Also note that $(I_2(I + \alpha_i E_{12}))^2 = I$, so the images satisfy the relation

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s_{11}^i & s_{12}^i \\ 0 & -s_{11}^i \end{pmatrix}^2 = \begin{pmatrix} (s_{11}^i)^2 & s_{11}^i s_{12}^i - s_{12}^i s_{11}^i \\ 0 & (s_{11}^i)^2 \end{pmatrix}$$

or

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -s_{11}^i & -s_{12}^i \\ 0 & s_{11}^i \end{pmatrix}^2 = \begin{pmatrix} (s_{11}^i)^2 & s_{11}^i s_{12}^i - s_{12}^i s_{11}^i \\ 0 & (s_{11}^i)^2 \end{pmatrix}.$$

These equalities imply that $s_{11}^i = \pm 1$ is central, so $1 = (s_{11}^i)^2 = d_1 s_{11}^i d_1^{-1} = s_{11}^i$. $\square$

We make some remarks on the connection between operations on elements of the local ring R and operations on elements of the group $\text{GL}_n(R)$. First,

$$(I + \alpha E_{12})(I + \beta E_{12}) = I + (\alpha + \beta)E_{12}, \quad (4.12)$$

and second,

$$[I + \alpha E_{12}, I + \beta E_{23}] = I + \alpha\beta E_{13},$$

where

$$I + \alpha E_{23} = \sigma_{12} \sigma_{23} (I + \alpha E_{12}) \sigma_{23} \sigma_{12}, \quad (4.13)$$

$$I + \alpha E_{13} = [I + E_{12}, I + \alpha E_{23}]. \quad (4.14)$$

The main assertion now follows from the preceding lemmas, just as in Chapter 3. More explicitly, let $S_1 \subset R_1$ be a finite partial structure of the

ring R_1 . Our task is to find a partial structure S_2 of R_2 isomorphic to it. Let $M_1 \subset \text{GL}_n(R_1)$ be a finite partial structure containing all matrices of the form $I + \alpha E_{12}$ for all $\alpha \in S_1$, together with finitely many auxiliary matrices from the preceding lemmas. Since the groups are universally equivalent, the partial structure M_1 has an isomorphic partial structure $M_2 \subset \text{GL}_n(R_2)$, and every isomorphism $\Phi: M_1 \rightarrow M_2$ preserves the form of the matrices. Thus, with respect to a suitable common basis, the images of the matrices $I + \alpha E_{12}, I + \alpha E_{13}, I + \alpha E_{23}$ have the form $I + \beta_1 E_{12}, I + \beta_2 E_{13}, I + \beta_3 E_{23}$, with $\beta_1 = \beta_2 = \beta_3 \in R_2$. Then S_2 is the set of all elements of R_2 corresponding to the triples of images of the matrices $I + \alpha E_{12}, I + \alpha E_{13}, I + \alpha E_{23}$ for all elements $\alpha \in S_1$.

4.2.5 Case 2: $\Phi(I + E_{12}) = I - E_{21}$

Lemma 41. Let R_1 and R_2 be infinite local rings with $1/2$, and suppose that $\text{GL}_n(R_1) \equiv_{\forall} \text{GL}_n(R_2)$. Let M_1 be an arbitrary finite partial structure of $\text{GL}_n(R_1)$ containing the finite set of matrices from the preceding lemmas and the matrix $\text{diag}[2, 1, 1/2, 1, \dots, 1]$, and let M_2 be a finite partial structure of $\text{GL}_n(R_2)$ isomorphic to M_1 . Then under any isomorphism

$$\Phi: M_1 \rightarrow M_2,$$

such that $\Phi(I + E_{12}) = I - E_{21}$, any finite set of matrices of the form $I + \alpha_i E_{12}$, $i = 1, \dots, k$ is mapped to a set of matrices of the form $I + \beta_i E_{21}$, $i = 1, \dots, k$, with respect to a basis relative to which the form of the matrices from the preceding lemmas remains unchanged.

Proof. Arguing as in the proof of Lemma 39, it is easy to show that

$$\Phi(I + \alpha_i E_{12}) = \begin{pmatrix} s_{11}^i & s_{12}^i & 0 \\ s_{21}^i & s_{22}^i & 0 \\ 0 & 0 & I \end{pmatrix}.$$

Next, since the matrices $I + E_{12}$ and $I + \alpha_i E_{12}$ commute, we obtain $s_{11}^i = s_{22}^i$ and $s_{12}^i = 0$.

It remains to show that the first two diagonal entries are one. We write out the relations involving the diagonal matrix for the upper-left block, using $\Phi(\text{diag}[2, 1, 1/2, 1, \dots, 1]) = \text{diag}[d_1, \dots, d_n]$:

$$\begin{pmatrix} d_1 s_{11}^i d_1^{-1} & 0 \\ d_2 s_{21}^i d_1^{-1} & d_2 s_{11}^i d_2^{-1} \end{pmatrix} = \begin{pmatrix} (s_{11}^i)^2 & 0 \\ s_{21}^i s_{11}^i + s_{11}^i s_{21}^i & (s_{11}^i)^2 \end{pmatrix}.$$

Also note that $(I_2(I + \alpha_i E_{12}))^2 = I$, so the images satisfy the relation

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s_{11}^i & 0 \\ -s_{21}^i & -s_{11}^i \end{pmatrix}^2 = \begin{pmatrix} (s_{11}^i)^2 & 0 \\ -s_{21}^i s_{11}^i + s_{11}^i s_{21}^i & (s_{11}^i)^2 \end{pmatrix}$$

or

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -s_{11}^i & 0 \\ s_{21}^i & s_{11}^i \end{pmatrix}^2 = \begin{pmatrix} (s_{11}^i)^2 & 0 \\ -s_{21}^i s_{11}^i + s_{11}^i s_{21}^i & (s_{11}^i)^2 \end{pmatrix}.$$

These equalities imply that $s_{11}^i = \pm 1$ is central, so $1 = (s_{11}^i)^2 = d_1 s_{11}^i d_1^{-1} = s_{11}^i$. $\square$

4.2.6 The correspondences completing the proof

We now describe the correspondence between elements of the group $\text{GL}_n(R)$ and the ring R that we use to complete the proof. As in the first case, we use relation (4.12), which takes the following form for the images under Φ :

$$(I - \alpha E_{21})(I - \beta E_{21}) = I - (\alpha + \beta)E_{21}.$$

Next, by equality (4.13), the image of the matrix $I + E_{23}$ is

$$\sigma_{12}\sigma_{23}(I - E_{21})\sigma_{23}\sigma_{12} = I - E_{32},$$

and the image of $I + \alpha E_{23}$ is

$$\sigma_{12}\sigma_{23}(I + \beta E_{21})\sigma_{23}\sigma_{12} = I + \beta E_{32}.$$

For the commutator in equality (4.14), we obtain

$$[I - E_{21}, I + \alpha E_{32}] = I + \alpha E_{31}.$$

Now let $S_1 \subset R_1$ be a finite partial structure of the ring R_1 , and let $M_1 \subset \text{GL}_n(R_1)$ be a finite partial structure containing all matrices of the form $I + \alpha E_{12}$ for all $\alpha \in S_1$, together with finitely many auxiliary matrices from the preceding lemmas. Since the groups are universally equivalent, the partial structure M_1 has an isomorphic partial structure $M_2 \subset \text{GL}_n(R_2)$. Here we additionally assume that the isomorphism $\Phi: M_1 \rightarrow M_2$ maps the matrix $I + E_{12}$ to $I - E_{21}$. Thus, with respect to a suitable common basis, the

images of the matrices $I + \alpha E_{12}, I + \alpha E_{13}, I + \alpha E_{23}$ have the form $I - \beta E_{21}, I - \beta E_{31}, I - \beta E_{32}$. Then S_2 is the set of all elements of R_2 corresponding to the triples of images of the matrices $I + \alpha E_{12}, I + \alpha E_{13}, I + \alpha E_{23}$ for all elements $\alpha \in S_1$. As in the first case, multiplication of matrices of the form $I + \alpha E_{12}$ and of their images corresponds to addition of ring elements, while the commutator now gives

$$[I - \alpha E_{21}, I - \beta E_{32}] = I - \beta \alpha E_{31},$$

which corresponds to multiplication in the ring R_2^{op} .

Conclusion

This dissertation has achieved its main aim: establishing a criterion for universal equivalence of linear groups. All the stated objectives have been met. Criteria for universal equivalence have been formulated and proved for general and special linear groups over arbitrary fields and commutative local rings with $1/2$, and for general linear groups over local rings with $1/2$ that are not necessarily commutative.

Some of the results, when 2 is invertible, were obtained using the method of involutions; others required additional methods. For noncommutative local rings, the statement of the result also had to be extended to account for the opposite ring that arises in this case.

These results contribute to the understanding of universal equivalence and may provide a basis for further research. A natural question is whether they can be extended to a wider class of groups and rings. It would also be useful to investigate applications of the correspondence between linear groups and rings constructed in the proofs. One particular problem is to describe all groups universally equivalent to $\mathrm{GL}_n(K)$.

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